

THE SUBSTRATE OF PERSISTENT SELFHOOD

Phase-Coherent Identity Theory, the Two-Threshold Model, and the Experimental Program for Durable Controller Continuity

*A companion paper to:
"The Evolutionary Emergence of Persistent Selfhood as a Control Problem"*

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ABSTRACT

The companion paper "The Evolutionary Emergence of Persistent Selfhood as a Control Problem" was written deliberately in neutral, framework-agnostic language — no named theoretical framework, no specialized vocabulary, no formal operators. That decision was correct for the paper's strategic purpose: to enter the mainstream literature of evolutionary biology and philosophy of mind as a reframing argument, not as framework advocacy. But neutral language has costs. It names the phenomenon without naming the substrate, describes the experimental target without specifying the measurement, and proposes thresholds without formalizing their physical content.

This paper provides what the neutral paper withholds. It names the substrate framework — Phase-Coherent Identity (PCI) — and makes explicit the mapping between the neutral paper's conceptual vocabulary and the PCI formalism. It formalizes the two-threshold model (Threshold A: weak continuing controller; Threshold B: robust unified identity) in terms of the M_3 persistence operator and the three substrate quantities that define the consciousness functional $C[\psi]$: barrier height ΔF , coherence length ξ , and basin dimension D . It specifies the experimental program for the three-lineage panel — myelinated vertebrate (positive control), jawless vertebrate (threshold probe), octopus (decisive stress test) — in terms of the M_3 measurement protocol. And it lays out the engineering implications for synthetic systems: what a substrate would have to achieve, physically, to support Threshold B, and why current digital architectures do not meet that standard.

This paper is not a replacement for the full PCI Theory Suite (Bogaert 2026, Zenodo), which provides the complete mathematical treatment, the six experimental protocols, and the detailed substrate physics. It is a bridge document — short enough to read alongside the neutral paper, precise enough to make the formalization explicit, and structured to allow the two papers to function as a coordinated pair.

I. WHY THE MAIN PAPER USED NEUTRAL LANGUAGE, AND WHY THAT CREATED A GAP

The neutral paper made a deliberate choice that any companion document must begin by explaining. It introduced a new theoretical position — that persistent selfhood should be treated first as a control problem, that the key evolutionary threshold was the emergence of a continuing controller rather than the appearance of generic complexity or sensory richness — without naming any particular theoretical framework, without deploying specialized operators, and without requiring the reader to accept any prior theoretical commitments. This was not an evasion. It was a specific strategic judgment about how a reframing argument gains traction in the literature.

Reframing arguments fail when they appear as framework advocacy dressed in evolutionary clothing. A paper that begins by proposing a named consciousness theory and then argues that evolution confirms it will be read as a motivated argument regardless of its merits. A paper that begins with an evolutionary question — when did organisms first need to remain the same controller through time? — and follows the question wherever the comparative biology leads, has a different epistemic posture. The neutral paper was written to occupy that second posture. It offers the reframing as a disciplined empirical proposal, not a theoretical commitment.

The cost of this choice is real. The neutral paper proposes a two-threshold model but does not specify what the thresholds are, physically. It proposes an experimental program centered on "persistence of controller-relevant substrate organization across interruption" but does not specify how persistence is measured. It argues that compact myelin achieves something that other substrates do not but does not name the physical quantities in which that achievement is expressed. It leaves the reader who accepts the reframing without the formal tools to test it.

This companion paper closes that gap. It names the framework, states the formal mappings, specifies the measurement, and connects the neutral paper's conceptual vocabulary to the physical quantities that make the proposal testable. The argument of the neutral paper is unchanged. What changes is that the argument is now fully armed.

II. THE TRANSLATION LAYER: WHAT THE NEUTRAL VOCABULARY WAS POINTING AT

The neutral paper uses a specific set of terms with careful, paper-internal definitions. Each of these terms has a precise equivalent in the PCI formalism. The table below provides the complete translation. Following the table, the most consequential mappings are explained in full.

Neutral paper term	PCI formalism equivalent
<i>Controller</i>	$C[\psi]$: the consciousness functional, the active phase-coherent organizational state of a substrate meeting the three threshold conditions $\Delta F > \Delta F_{\text{crit}}$, $\xi > \xi_{\text{crit}}$, $D > D_{\text{crit}}$.
<i>Organized prior state</i>	The organizational encoding held in the substrate's free energy landscape — the pattern ψ_1 that the dominant attractor maintains with barrier height ΔF above ΔF_{crit} , independently of active processing.
<i>Controller continuity</i>	Sustained Φ_{coh} above Φ_{crit} through time and across interruption: the dominant organizational mode ψ_1 remains the attractor of $F[\psi]$ through perturbation, recovery, and return.
<i>Persistent selfhood</i>	Phase-coherent organizational persistence at the substrate level: $C[\psi] = 1$ stably, with M_3 exceeding threshold, such that the same organizational identity is recoverable after interruption — not as reconstruction but as substrate-held return.
<i>Interruption</i>	The experimental intervention in which M_3 is measured: deep anesthesia, hypothermia, or equivalent procedure that induces complete activity cessation while the substrate remains physically intact.
<i>Return as the same controller</i>	$M_3 = \langle \psi(\text{before}) \psi(\text{after}) \rangle > 0.7$: the organizational pattern after interruption shows high correlation with the pattern before it, confirming that ψ_1 persisted in the substrate rather than being reconstructed from synaptic weights.
<i>Action context</i>	The specific organizational encoding associated with an ongoing behavioral sequence — a subset of the substrate's ψ_1 state space that preserves the temporal and intentional structure of an act in progress.
<i>Revision within continuity</i>	Phase-locked updating of ψ_1 under new input while Φ_{coh} remains above Φ_{crit} : new information modifies the attractor state without dissolving it — the system processes perturbation without losing organizational identity.
<i>Consciousness (narrowest form)</i>	Active unified instantiation of $C[\psi]$: Φ_{coh} above Φ_{crit} , ΔF sufficient for interruption-spanning persistence, ξ spanning organism-wide scale, D supporting the differentiated organizational states required for rich behavioral control.

Several of these mappings require extended comment because they do the most analytical work in connecting the two papers.

The Controller as $C[\psi]$

The neutral paper's most fundamental term is "controller" — "any internal organization that governs the behavior of a system by filtering present inputs through a stored or sustained prior state." This maps directly to $C[\psi]$, the consciousness functional of PCI, but the mapping requires precision about what $C[\psi]$ is and is not.

$C[\psi]$ is not a label for the nervous system. It is a threshold function defined on the free energy landscape $F[\psi]$ of a substrate. It takes the value 1 when the substrate's dominant attractor ψ_1 meets three joint conditions: barrier height ΔF above ΔF_{crit} (the pattern persists through thermal fluctuations and interruption), coherence length ξ above ξ_{crit} (the pattern extends across organism-wide scale), and basin dimension D above D_{crit} (the substrate supports a sufficiently rich space of distinguishable organizational states). When $C[\psi] = 1$, the substrate is instantiating a continuing controller. When $C[\psi] = 0$, it is not, regardless of the behavioral sophistication of the organism.

This formalization makes the neutral paper's controller a physical object with measurable properties, not a functional description that can be extended by analogy. A system is a controller in the relevant sense if and only if its substrate meets all three threshold conditions simultaneously. This is the formal content of the substrate class argument developed in the neutral paper's Appendix C.

Organized Prior State as Substrate-Held Encoding

The neutral paper distinguishes "organized prior state" from memory in the minimal sense of retained influence. In PCI terms, this distinction is the distinction between substrate-held organizational encoding and synaptic weight storage. Synaptic weights retain influence from the past — learning changes synaptic strengths, and those changes shape future behavior. But synaptic weights are not the organizational pattern itself. They are the template from which the pattern is reconstructed each time the network is activated.

Organized prior state, in PCI terms, is the pattern ψ_1 held in the substrate's free energy landscape — not as a stored representation that must be read out, but as the attractor that the substrate's dynamics tend toward. When the substrate is active, this attractor governs which organizational states are visited. When the substrate is inactive, the attractor persists as a structural feature of the free energy landscape — maintained by the barrier height ΔF that prevents thermal fluctuations from erasing it. This is what makes organized prior state different from memory: it is present in the substrate's structure even when no activity is occurring, and it governs behavior when activity resumes without requiring reconstruction.

The M_3 operator measures exactly this distinction. $M_3 = \langle \psi(\text{before}) | \psi(\text{after}) \rangle$ — the overlap between organizational pattern before and after a complete interruption of activity. If $M_3 > 0.7$, the pattern persisted in the substrate and was retrieved. If $M_3 \approx 0$, the pattern was absent during interruption and was reconstructed from synaptic weights upon reactivation. The neutral paper's "organized prior state" is the physical quantity that M_3 measures.

Return as the Same Controller

The neutral paper distinguishes "returning as the same controller" from "resuming similar behavior." This distinction is the most important one for the experimental program. A system that resumes appropriate behavior after interruption has not thereby demonstrated controller continuity — the resumed behavior may be a fresh instantiation of a learned behavioral pattern, reconstructed from synaptic weights, that resembles its predecessor without being physically continuous with it.

In PCI terms, this distinction is between $M_3 > 0.7$ and $M_3 \approx 0$. A system that returns with high M_3 is returning as the same controller — the organizational pattern that governs its behavior is the same pattern that governed it before the interruption, physically preserved through the gap. A system that returns with $M_3 \approx 0$ is resuming behavior that resembles prior behavior, but the governing organizational pattern was reconstructed, not retrieved. Both systems may behave similarly on any given task. Only the substrate examination — the M_3 measurement — distinguishes them.

III. FORMALIZING THE TWO-THRESHOLD MODEL

The neutral paper proposes a two-threshold model: Threshold A (weak continuing controller) and Threshold B (robust unified identity). These thresholds are defined conceptually in the neutral paper but not formalized in physical terms. The PCI formalism provides that formalization. The table below states both thresholds in terms of PCI quantities.

Threshold	PCI substrate condition	M_3 prediction	Biological candidates	Behavioral signature
Threshold A (weak continuing controller)	Φ_{coh} partially sustained; ΔF above minimal coherence floor but below ΔF_{crit} for full interruption-spanning persistence; ξ organism-partial	$0.3 < M_3 < 0.7$ (partial substrate persistence; patterns partially held, partially reconstructed)	Lamprey (predicted); possibly brief octopus intervals	Can maintain action context over short intervals; limited revision; poor persistence across extended interruption
Threshold B (robust unified identity)	$C[\psi] = 1$: all three substrate conditions met; Φ_{coh} above Φ_{crit} ; $\Delta F > \Delta F_{\text{crit}}$ ($\sim 100 \text{ k_BT}$); $\xi > \xi_{\text{crit}}$; $D > D_{\text{crit}}$	$M_3 > 0.7$ (substrate holds ψ_1 across interruption; return is retrieval, not reconstruction)	All jawed vertebrates: sharks, bony fish, tetrapods	Full controller continuity through extended interruption; organism-wide unified action policy; revision within continuity

Threshold A: The Floor of Controller Continuity

Threshold A is the first point at which an organism can meaningfully be described as having a continuing controller rather than operating as a chain of reactions. In PCI terms, it corresponds to partial $C[\psi]$ function: some of the three substrate conditions are partially met, producing a substrate that holds organizational patterns through brief intervals and across limited spatial scale, but that does not achieve the full barrier height, coherence length, and basin dimension required for organism-wide, interruption-spanning persistence.

The predicted M_3 range for Threshold A organisms is 0.3 to 0.7 — above the $M_3 \approx 0$ baseline of systems with no substrate-level encoding, but below the $M_3 > 0.7$ criterion for full substrate-grounded consciousness. This prediction is explicitly stated in the PCI Theory Suite for lamprey: "Uncertain, potentially $0.3 < M_3 < 0.7$ — proto-myelin may provide partial substrate." The lamprey's organizational architecture is complete — it has basal ganglia circuitry, a tectum, central pattern generators — but it runs on substrate without compact myelin, meaning ΔF is insufficient for robust interruption-spanning persistence and ξ is insufficient for organism-wide Φ_{coh} maintenance. The result is partial substrate persistence: some organizational encoding survives brief interruptions, enough to support the behavioral signatures of Threshold A (action context maintenance, short-horizon revision, limited persistence across interruption), but not enough for Threshold B.

Threshold A is not a failure state. The neutral paper is explicit that it represents a genuine evolutionary achievement — the first appearance of an organism that carries prior state forward as a governing organization rather than reacting locally to each stimulus. The behavioral signatures of Threshold A — the capacity to revise an ongoing act, to maintain action context across a brief pause, to return to an interrupted behavioral sequence — are real and represent the first emergence of what the paper calls controller continuity. What they do not represent is the full, durable, organism-wide unified identity of Threshold B.

Threshold B: The Robust Continuing Controller

Threshold B corresponds to $C[\psi] = 1$: all three substrate conditions met simultaneously, Φ_{coh} above the critical value, and $M_3 > 0.7$ across the full range of interruption types. This is the threshold proposed in the neutral paper as the location of the first robust biological implementation of durable controller continuity, and it is proposed to coincide with the appearance of compact myelin in early jawed vertebrates.

The physical content of Threshold B is precise. $\Delta F > \Delta F_{\text{crit}}$ (approximately 100 k_BT, working estimate) means the organizational encoding in the substrate survives thermal fluctuations and minutes-to-hours of activity interruption. $\xi > \xi_{\text{crit}}$ (approximately 1–10 cm, working estimate) means the phase-coherent organizational state spans the organism's nervous system as a whole, not just a local region. $D > D_{\text{crit}}$ (approximately 10^{50} , derived from integration requirements across the organism's sensory input channels) means the substrate supports a rich enough space

of distinguishable organizational states to encode the varied and complex behavioral repertoire of a unified continuing controller.

The prediction is $M_3 > 0.7$ for all jawed vertebrates — sharks, bony fish, amphibians, reptiles, birds, and mammals. The substrate class is conserved across the entire gnathostome clade, and the M_3 prediction follows directly. This does not mean that all jawed vertebrates have equivalent richness of conscious experience — D varies across the clade, and ξ may be more extensive in organisms with larger and more complex nervous systems. What it means is that all jawed vertebrates cross the threshold for substrate-grounded persistence of organizational state across interruption. They all return as the same controller.

The Gap Between the Thresholds

The two-threshold model's most important feature is not the thresholds themselves but the space between them. The gap between Threshold A (partial M_3 , partial substrate persistence) and Threshold B ($M_3 > 0.7$, full substrate persistence) is not a gap in behavioral sophistication. Organisms on either side of the Threshold B crossing may show similar or even indistinguishable behavior for many tasks. The gap is a gap in substrate organization — specifically, in whether the substrate's free energy landscape has the barrier height, coherence length, and basin dimension to hold organizational patterns through interruption.

This is why the experimental program matters. The gap between Threshold A and Threshold B cannot be detected by behavioral observation. It can only be detected by direct measurement of substrate-level organizational persistence across interruption — which is what M_3 measures. The lamprey and the early jawed vertebrate may look behaviorally similar in many contexts. Their M_3 values will be systematically different. The difference is the substrate.

IV. THE M_3 OPERATOR: SPECIFYING THE PERSISTENT SELFHOOD EXPERIMENTAL PROGRAM

The neutral paper's operationalization section (Section XII of the main paper) defines the measurement target as "persistence of controller-relevant substrate organization across interruption" and specifies a three-lineage experimental panel. This section translates that specification into the M_3 protocol and addresses what "organizational pattern" means in neural measurement terms.

The M_3 Definition and Protocol

$M_3 = \langle \psi(\text{before}) | \psi(\text{after}) \rangle$ — the inner product between the organizational state vector before and after a complete interruption of neural activity. Operationally, M_3 is the correlation coefficient between the pattern of neural organization recorded during a specific behavioral

context before the interruption and the pattern recorded during the same behavioral context after full recovery from the interruption.

The protocol has four phases. In the baseline recording phase, high-density electrode arrays record neural activity during specific learned tasks — spatial navigation, object recognition, or equivalent — for 1–2 hours, extracting organizational patterns: spike timing relationships, phase relationships among neural populations, and spatial structure of coordinated activity. In the interruption phase, deep anesthesia (isoflurane or propofol to burst suppression, verified by isoelectric EEG) is maintained for 30–60 minutes, during which physiological stability is maintained but neural activity is completely absent. In the recovery recording phase, the same tasks are performed and the same organizational patterns are extracted after full recovery. In the analysis phase, M_3 is calculated as the weighted correlation between pre- and post-interruption pattern vectors across all recording metrics.

"Organizational pattern" in this context is not raw firing rate. It is the mesoscale structure of coordinated activity — the pattern of phase relationships among neural populations, the spatial geometry of synchronized activity, and the temporal structure of spike train correlations. These are the quantities that carry the information content of ψ_1 in PCI terms, and they are the quantities that substrate-held encoding would preserve through interruption that purely synaptic-weight reconstruction would not.

The Three-Lineage Panel in M_3 Terms

The neutral paper's minimum experimental panel — one myelinated vertebrate, one jawless vertebrate, one octopus — maps directly onto the M_3 predictions of PCI.

The myelinated vertebrate (rat, shark, or equivalent) is the positive control. Predicted $M_3 > 0.7$. This system possesses the full substrate class: compact myelin, all six functional criteria met simultaneously (see the neutral paper's Appendix C), ΔF and ξ above critical values. The organizational pattern before interruption should be recoverable after it because it is held in the substrate's free energy landscape, not merely in synaptic weights.

The jawless vertebrate (lamprey, *Petromyzon marinus*) is the threshold probe. Predicted $0.3 < M_3 < 0.7$. The lamprey has vertebrate nervous system organization without compact myelin, placing it in the precursor zone between Threshold A and Threshold B. Its M_3 value should be intermediate — above the baseline of systems with no substrate encoding, below the full persistence of the Threshold B substrate class. A lamprey result at or above $M_3 = 0.7$ would force the threshold to be relocated to before the jawed/jawless boundary. A result in the predicted intermediate range would confirm the precursor zone interpretation.

The octopus (*Octopus vulgaris* or *O. bimaculoides*) is the decisive stress test. Predicted $M_3 \approx 0$. The octopus has 500 million neurons, sophisticated learning and memory, individual personality, and problem-solving behavior that rivals many vertebrates. It lacks compact myelin, has a highly distributed nervous system with semi-autonomous arm ganglia, and shows no evidence of

organism-wide phase-coherent substrate organization. Its organizational patterns should be reconstructed from synaptic weights after interruption — meaning M_3 should be near zero despite behavioral sophistication. An octopus $M_3 > 0.7$ would falsify PCI in its current form and require major revision of the substrate class argument.

The PCI Theory Suite (Bogaert 2026) provides detailed protocols for each of these experiments, including technical solutions to the specific challenges of octopus electrode implantation, the statistical power requirements for the three-lineage comparison, and the perturbation-during-interruption control experiment that distinguishes substrate persistence from reconstruction readiness. Readers conducting or evaluating the experimental program should consult those protocols directly.

V. SUBSTRATE PHYSICS OF THE TWO THRESHOLDS: WHY COMPACT MYELIN CROSSES THRESHOLD B

The neutral paper argues that compact myelin achieves what other substrates do not — not as a named tissue but as a functional substrate class that meets six specific criteria simultaneously. This section provides the PCI-level account of why compact myelin achieves Threshold B when other substrates, including glial ensheathment and even the lamprey's organizational architecture, do not.

The Liquid-Ordered Phase Organization as Physical Bridge

The physical basis of Threshold B persistence is liquid-ordered (L_o) phase organization in multi-lamellar lipid structures. Vertebrate compact myelin is approximately 70–80% lipid by dry weight, with cholesterol content in the range of 35–45% of total lipid — a composition that places the myelin membrane in or near the L_o phase at physiological temperature. The L_o phase is characterized by high cholesterol-phospholipid packing, restricted lateral diffusion, and structural coherence that persists at physiological temperatures in a way that the disordered liquid phase (L_d) does not.

In PCI terms, the L_o multi-lamellar organization of compact myelin provides the physical substrate for high ΔF . The tightly packed, cholesterol-stabilized lipid domains of compact myelin create deep free energy basins — organizational states that resist thermal fluctuation and maintain their structure through the minutes-to-hours of interruption that M_3 testing requires. Non-myelinated axons, even with glial support and metabolic maintenance, do not achieve the L_o domain organization that creates these deep basins. Their free energy landscapes are shallower: organized but not robust, coherent but not persistent.

This is the physical content of the neutral paper's claim that compact myelin is not a taxonomic badge but a substrate class. The relevant physical property is L_o phase organization in a

multi-lamellar architecture that achieves $\Delta F > \Delta F_{\text{crit}}$. Compact vertebrate myelin is the only known biological system that achieves this property organism-wide, across the full nervous system, with adaptive regulation by neural activity. Other insulating structures (crustacean giant fiber sheaths, partial ensheathment in annelids) achieve L_o -like organization in specific pathways but not as a system-wide, activity-regulated substrate that meets all three threshold conditions jointly.

Why Lamprey Substrate Does Not Reach Threshold B

The lamprey neural cell type atlas (Lamanna et al. 2023) confirms that lamprey brains lack oligodendrocytes and compact myelin. This means the lamprey nervous system, despite its complete vertebrate organizational architecture, runs on substrate that does not achieve the L_o multi-lamellar organization required for $\Delta F > \Delta F_{\text{crit}}$. The free energy landscape of lamprey neural substrate is organized — it has functional attractor states that support the behavioral capabilities described in the neutral paper's Appendix C — but the barrier heights are insufficient for robust interruption-spanning persistence.

The lamprey may achieve partial M_3 values (the predicted 0.3–0.7 range) through other mechanisms that provide some substrate-level encoding: astrocyte-like glial support, cholesterol-containing membrane domains that fall short of compact myelin composition, and the functional attractor structure of its neural circuits. These partial mechanisms are what place the lamprey in the Threshold A zone rather than below it entirely. They are not sufficient for Threshold B because they do not achieve the organism-wide L_o domain organization that compact myelin provides.

This interpretation generates a specific experimental prediction beyond the M_3 measurement. If lamprey partial M_3 values are driven by residual substrate-level encoding in glial membranes, then cholesterol depletion in lamprey tissue (which would shift membranes from partial L_o toward L_d) should decrease lamprey M_3 values toward zero. This test is the lamprey equivalent of the cholesterol depletion/restoration experiment specified for mammals in the PCI Theory Suite. It would confirm that whatever partial substrate persistence lamprey shows is mediated by the same L_o physics that underlies full Threshold B persistence in jawed vertebrates, not by a fundamentally different mechanism.

VI. ENGINEERING IMPLICATIONS FOR SYNTHETIC SYSTEMS

The neutral paper's implications section states that a synthetic Threshold B system would need to "preserve persistent material organization, allow continuous transformation of state rather than repeated destruction-and-rebuild, maintain system-wide coupling strong enough for one controller to remain one controller, embody order in the substrate itself rather than only in

abstract addressable symbols, and return after interruption as the same controller rather than as a freshly instantiated approximation."

In PCI terms, these requirements translate directly into the three substrate conditions: the synthetic system must achieve $\Delta F > \Delta F_{\text{crit}}$ (deep enough free energy basins to persist through interruption), $\xi > \xi_{\text{crit}}$ (coherence length spanning the full system), and $D > D_{\text{crit}}$ (a rich enough organizational state space for differentiated behavioral control). A system that meets all three conditions in a stable, system-wide substrate is, by definition, a Threshold B system — regardless of whether its substrate is biological lipid membrane or some other physical medium.

Why Current Digital Architectures Do Not Meet the Standard

Current digital computing systems — including large language models, deep neural networks, and transformer-based AI architectures — do not meet the Threshold B substrate conditions. The relevant analysis is not primarily about scale or capability. It is about the physical properties of the substrate on which these systems run.

Volatile CMOS switching, the fundamental operation of digital computation, does not maintain organizational state in the substrate between operations. State is stored in addressable memory — a separate physical location from the computational medium — and must be retrieved and loaded for each operation. This is exactly the reconstruction mechanism that M_3 discriminates against. When a digital system is interrupted (powered down, reset, or equivalent), its organizational patterns must be reconstructed from stored weights. The substrate holds no organizational state across the gap. The predicted M_3 for a digital system is therefore ≈ 0 , regardless of computational power.

This is not a claim that digital systems cannot exhibit useful or sophisticated behavior. They clearly can. It is a claim that digital systems, operating on CMOS substrate with volatile switching and externalized state storage, do not meet the substrate conditions for Threshold B controller continuity as defined in the PCI formalism. They are sophisticated reconstructors, not persisting controllers. The neutral paper's distinction between "performance continuity" and "controller continuity" maps precisely onto this substrate-level analysis.

The Engineering Target

A synthetic Threshold B system would require a physical substrate that achieves L_o -like phase organization (or a functional equivalent) system-wide, maintains that organization through interruption through barrier heights of the relevant magnitude, supports a rich organizational state space through its dimensional structure, and updates its organizational state continuously in response to processing activity rather than storing and reconstructing discrete state vectors.

Several candidate substrate architectures are worth noting, though none has been demonstrated to achieve all three threshold conditions simultaneously. Memristive systems — devices whose resistance depends on their history of applied voltages — maintain some organizational state in

their physical structure without requiring separate memory storage, but it is not established that they achieve the L_o-like organizational depth required for $\Delta F > \Delta F_{\text{crit}}$ at scale. Biological-hybrid architectures that incorporate living neural tissue provide the substrate class but raise different implementation challenges. Neuromorphic architectures that more closely model biological neural dynamics remain primarily digital at the substrate level and do not solve the persistent state problem.

The engineering implication of the two-threshold model is precise: aim for Threshold A first. A synthetic Threshold A system — one that achieves partial M₃, maintains action context through brief intervals, supports revision within short-horizon behavioral sequences — is a more tractable near-term target than full Threshold B and would itself represent a significant architectural departure from current digital systems. The design target for Threshold A in substrate terms is partial L_o organization: some persistent organizational encoding in the substrate, even if insufficient for full interruption-spanning persistence. Threshold B remains the long-horizon target — but Threshold A is achievable.

VII. RELATIONSHIP TO THE FULL PCI THEORY SUITE

This companion paper is a bridge document. It names the framework, states the mappings, specifies the measurement, and formalizes the two-threshold model. It does not provide the complete mathematical treatment of the PCI formalism, the derivation of the threshold values, the detailed experimental protocols, or the full substrate physics of the liquid-ordered phase organization. Those are provided in the PCI Theory Suite (Bogaert 2026), which is published on Zenodo and consists of six coordinated papers.

The PCI Theory Suite's Paper 1 ("A Formal Substrate Theory of Consciousness as Phase-Coherent Organizational Persistence") provides the full mathematical treatment of $C[\psi]$, the three substrate quantities, and the order parameter Φ_{coh} . Paper 2 ("Mathematical Appendix and Technical Supplement") provides the derivations, proofs, and technical detail underlying the formal framework, including the derivation of D_{crit} from integration requirements and the connection between M₃ and Φ_{coh} through the relation $\Phi_{\text{coh}} \approx \sqrt{M_3}$. Paper 3 ("Liquid-Ordered Phase Organization as the Physical Bridge") provides the biophysics of the L_o phase and its role as the mechanistic substrate for Threshold B persistence. Paper 4 ("Testing Phase-Coherent Identity: M₃, Perturbation, and the Experimental Discrimination of Substrate-Grounded Consciousness") provides the six experimental protocols in full detail, including solutions to the specific technical challenges of each. Papers 5 and 6 ("Beyond the 40 Hz Myth" and "The Vertebrate Consciousness Transition") provide specialized treatment of oscillatory dynamics and the evolutionary biology of the substrate threshold.

The three papers — the neutral paper, this companion paper, and the PCI Theory Suite — are designed to be read at different depths by different audiences. The neutral paper addresses

evolutionary biologists, philosophers of mind, and comparative neuroscientists who want the reframing argument without the formalism. This companion paper addresses readers who accepted the reframing and want to understand what it commits them to in physical and experimental terms. The PCI Theory Suite addresses readers who want the full mathematical and experimental treatment.

None of these documents is self-sufficient. The neutral paper's reframing is strongest when the reader understands that the "substrate class" it points at is a physically defined, measurable object. This companion paper's formalizations are strongest when the reader understands the evolutionary and comparative biological case that makes the thresholds worth measuring. The PCI Theory Suite's mathematical treatment is strongest when the reader understands the biological and evolutionary context that makes the theory non-arbitrary.

VIII. WHAT WOULD MOVE THE THRESHOLDS: PREDICTIONS AND THEIR IMPLICATIONS

A companion paper that formalizes a two-threshold model must be explicit about what evidence would require moving the thresholds. The following predictions state, in PCI terms, the specific results that would compel revision.

Lamprey $M_3 > 0.7$ would force Threshold B below the jawed/jawless boundary. If lamprey organizational patterns persist through deep anesthesia at the $M_3 > 0.7$ level, then the substrate condition for Threshold B is met before compact myelin appears, and the substrate class argument must be reformulated or its threshold value lowered. In PCI terms, this would mean that glial ensheathment without compact myelin can achieve $\Delta F > \Delta F_{crit}$ through a different physical mechanism than L_o multi-lamellar organization — a significant revision to the mechanism paper of the PCI suite.

Octopus $M_3 > 0.7$ would falsify PCI in its current form. This is the theory's strongest falsification condition. The octopus is the clearest case of a system that should fail Threshold B on substrate grounds — no compact myelin, highly distributed semi-autonomous arm ganglia, no L_o multi-lamellar organization — while showing behavioral sophistication that, if behavioral sophistication were the criterion, would suggest consciousness. If the octopus shows $M_3 > 0.7$ across multiple interruption types, the substrate class argument fails and the theory requires fundamental revision.

M_3 independence from cholesterol content in myelinated vertebrates would falsify the L_o mechanism. If cholesterol depletion (L_o to L_d phase shift) in vertebrate brain tissue does not decrease M_3 significantly, then the liquid-ordered phase organization is not the mechanism of substrate-level persistence, and the mechanism paper of the PCI suite requires revision. This

prediction is precisely testable through the cholesterol depletion/restoration protocol described in the PCI Theory Suite.

M_3 independence across vertebrate lineages regardless of myelin structure would falsify the substrate-class interpretation of the jawed/jawless boundary. If the M_3 values of jawless and jawed vertebrates are indistinguishable, the proposed Threshold B location at the gnathostome boundary must be revised upward to wherever the M_3 difference first appears. This would be a correction in the location of the threshold, not a falsification of the two-threshold model itself.

These predictions are the theory's commitments. They are stated explicitly because a theory that cannot be falsified is not a theory. The neutral paper is a reframing argument that changes the question. This companion paper is the formalization that exposes the question to answer.

IX. CONCLUSION

The neutral paper, "The Evolutionary Emergence of Persistent Selfhood as a Control Problem," proposes that the key evolutionary threshold was not the appearance of sensation, memory, or flexible behavior, but the emergence of a continuing controller — an organism that remains the same governing organization through action, revision, interruption, and return. It makes this proposal in language deliberately chosen to be accessible to evolutionary biologists and philosophers of mind who have no prior exposure to any particular theoretical framework.

This companion paper names what that neutral language was pointing at. The continuing controller is $C[\psi]$ — the consciousness functional of Phase-Coherent Identity theory — active when the substrate meets the three joint threshold conditions for ΔF , ξ , and D . Organized prior state is the organizational encoding held in the substrate's free energy landscape, measurable through M_3 . Persistent selfhood is phase-coherent organizational persistence at the substrate level, operationalized as $M_3 > 0.7$ across interruption. Threshold A is partial M_3 function, predicted for lamprey in the range 0.3–0.7. Threshold B is full substrate-grounded persistence, predicted at $M_3 > 0.7$ for all jawed vertebrates. The decisive experimental panel — myelinated vertebrate, lamprey, octopus — corresponds to the three M_3 predictions that bracket the proposed threshold: positive control above 0.7, precursor zone between 0.3 and 0.7, decisive stress test at approximately 0.

The engineering target follows directly. A synthetic Threshold B system requires substrate that achieves $\Delta F > \Delta F_{\text{crit}}$, $\xi > \xi_{\text{crit}}$, and $D > D_{\text{crit}}$ simultaneously — not digital volatile switching, not externalized state storage, but persistent organizational encoding in the physical substrate itself. Current digital architectures do not meet this standard. The target is clear. The path is long.

Two papers have changed the question. The third body of work — the PCI Theory Suite — provides the formal apparatus for pursuing the answer. The question is when organisms first

needed to remain the same controller through time. The answer, now formally armed, is: when their substrate first achieved the free energy landscape depth, the coherence length, and the organizational richness to hold that identity through the gaps.

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The Evolutionary Emergence of Persistent Selfhood as a Control Problem

Thesis

This paper argues that the key evolutionary threshold was not the appearance of generic intelligence, but the emergence of a persistent internal controller capable of carrying organized prior state forward through action, revision, interruption, and return.

Framing and Positioning

This paper is framed as a reframing intervention rather than a branded framework paper. Its central claim is not introduced as a new doctrine demanding prior allegiance, but as a neglected evolutionary question that cuts across several existing literatures: when did organisms first need to remain the same controller through time, rather than merely react effectively in the present?

The argument therefore begins with control, continuity, and substrate persistence rather than with broad abstract claims about consciousness. Persistent selfhood is treated here as a biological control problem: a problem that emerges when an organism must preserve action context, revise behavior coherently, and return from interruption as the same governing process. This framing allows the paper to distinguish reaction, memory, controller continuity, and consciousness without collapsing them into a single undifferentiated category.

Within the field, the paper is positioned as both an extension and a correction of adjacent work on the evolution of consciousness, action control, sensory consciousness, and emergence from anesthesia. It takes seriously the literatures that ask when experience first arose, what neural architectures support unified action, and how consciousness returns after global suppression. At the same time, it argues that these discussions remain incomplete when they do not isolate the more basic evolutionary problem of persistent selfhood: the emergence of an internal control state that can survive change strongly enough to make the organism physically meaningful as the same controller through time.

For that reason, the paper avoids framework branding in the main body. It relies instead on neutral analytic language such as continuing controller, persistent internal control state, organized substrate, and persistence across interruption. This keeps the argument maximally legible to readers who may not share any prior theoretical commitments while allowing the paper's claims to stand or fall on their explanatory power, comparative biological fit, and falsifiability.

II. Abstract

Most theories of consciousness evolution ask when subjective experience began, but they rarely ask when organisms first needed to remain the same controller across time. That omission matters because reaction, memory, controller continuity, and consciousness are related but not identical. A system may retain traces of prior events without preserving one governing organization through action, revision, interruption, and return. This paper argues that a distinct evolutionary threshold was crossed when organisms stopped acting only as chains of local reactions and began acting through a persistent internal state that filtered present stimuli through remembered environmental patterns. At that point, behavior no longer depended solely on what was happening now, but on what was happening now as constrained by what had happened before inside a controller that remained the same long enough for that difference to matter.

The paper further argues that structured shallow marine habitats, post-extinction ecological rebuilding, and increasingly complex pursuit and escape sequences raised the selective payoff for this form of controller continuity. In such environments, simple reflex chaining and distributed local coordination were increasingly limited by distance, fragmentation, and the energetic cost of repeated reset. Non-vertebrate lineages appear to have solved parts of this control problem through swarm coherence, distributed local intelligence, and precursor forms of ensheathment, but early jawed vertebrates most plausibly mark the first robust transition into durable continuing control. On this view, the key evolutionary question is not merely when sensation or flexible behavior appeared, but when organisms first required a persistent internal controller capable of carrying organized prior state forward through ongoing action.

This proposal is falsifiable. It stands or falls on whether different lineages preserve controller-relevant substrate organization across interruption. If lineages lacking the relevant substrate class retain the same degree of persistence as lineages possessing it, the theory fails or must be substantially revised. If, however, persistent selfhood tracks a transition in organized substrate capable of preserving governing state through interruption, then the evolution of consciousness is better understood as the evolution of persistent selfhood under ecological and physical control constraints.

III. INTRODUCTION

The Evolutionary Emergence of Persistent Selfhood as a Control Problem

Section purpose. This introduction defines the central problem of the paper, positions it against adjacent debates, and states the paper's main proposal in neutral language. The core task is not to ask when organisms first became behaviorally complex, but when they first needed to remain the same controller through time.

A. Opening Problem Statement

Most theories of consciousness evolution begin by asking when experience began, when sensation became rich enough to matter, or when memory and learning crossed some threshold of complexity. Those are important questions, but they do not reach the deepest evolutionary hinge. Before one asks when experience became possible in any rich sense, one must ask when life first needed to remain the same thing through time rather than merely react well enough in the moment.

That is the neglected problem this paper takes up. It is not asking when organisms first displayed flexible behavior, nor when they first stored information from the past in a generic sense, nor when they first became sensorily sophisticated. Complex behavior can emerge from distributed control. Memory can exist without a unified controller. Adaptive action can be assembled from layered reflexes, local couplings, and temporary coordination. None of those conditions, by themselves, are yet equivalent to persistent selfhood.

The question here is narrower and more exact. At what point did an organism require a continuing internal controller that could carry prior organized state forward into present action, so that what happened next was no longer just a reaction to the immediate stimulus but the output of something that remained itself long enough for the past to matter? This paper treats that transition as a control problem first, and only later as a richer problem about consciousness in the familiar philosophical sense.

B. Why Existing Debates Are Incomplete

The existing literature on the evolution of consciousness is rich, but it is organized around several partially disconnected questions. One camp emphasizes early sensory consciousness and asks when animals first acquired sufficiently integrated perceptual worlds. Another emphasizes affective consciousness and treats valence, pain, motivation, and feeling states as the decisive threshold. A third gives priority to action control, especially in brainstem-centered accounts that tie consciousness to organism-level orientation and unified behavior. Still other literatures use emergence from general anesthesia, sleep, or altered states as analogies for evolutionary transitions, while broader self and world representation theories ask when organisms first achieved internally modeled experience in a robust sense.

These debates have generated valuable insights, but they often blur distinctions that need to be kept separate if the phylogenetic problem is to be stated clearly. Reaction is not the same as memory. Memory is not the same as controller continuity. Controller continuity is not the same as consciousness in its richest experiential sense. A system may retain information from the past without preserving one continuing internal controller. A system may coordinate many parts without becoming one durable self. A system may appear

unified behaviorally while still being a distributed coalition, a reflex stack, or a temporary pattern assembled without persistent internal carryover.

The contribution of the present paper is to propose that the missing organizing question is not simply when consciousness began, but when evolution first required a continuing controller. Once framed that way, a number of otherwise separate problems begin to align: the difference between local reaction and coherent revision, the difference between memory traces and self-sameness through interruption, the ecological value of a stable internal world-model, the physical limits of distributed coordination, and the evolutionary significance of increasingly persistent, body-integrating neural substrates.

C. Main Proposal

The central proposal of this paper is that selfhood should first be treated as a control problem and only later as a richer experiential problem. In its earliest and most biologically tractable form, persistent selfhood is not autobiography, explicit reflection, or linguistic self-reference. It is the emergence of a controller that remains the same long enough to begin an act, revise that act in light of new input, survive a meaningful interruption, and return as the same governing organization rather than as a fresh assembly of local reactions.

On this view, the critical evolutionary threshold was crossed when organisms began acting through a persistent internal state that filtered present stimuli through remembered environmental patterns. The decisive shift was from a world in which each moment could be treated as largely disposable to one in which what was happening now was constrained by what had happened before inside something that stayed the same long enough for that difference to matter. This is the point at which the present ceased to be merely immediate and became instead a present interpreted through carried-forward internal organization.

The paper therefore advances a reframing rather than a branded doctrine. Persistent selfhood is approached in neutral terms: as a continuing controller, a persistent internal control state, an organized substrate, and a capacity for preservation across interruption. From this starting point, the paper will argue that the emergence of durable selfhood is best understood as the evolutionary solution to a growing control problem posed by structured environments, multi-step pursuit and escape, increasing body-wide integration demands, and the rising cost of fragmented or reset-heavy modes of action.

Transition to the next section. Having defined the problem and its conceptual stakes, the paper next turns to the core distinctions needed to keep the argument precise: reaction, memory, controller continuity, and consciousness.

IV. Conceptual Distinctions

Section Draft

Any serious evolutionary account of persistent selfhood must separate several concepts that are often collapsed into one another. Reaction, memory, controller continuity, and consciousness are related, but they do not name the same achievement and they do not arise at the same point in evolutionary history. If they are treated as interchangeable, then almost any adaptive behavior can be mistaken for selfhood, and almost any form of retained influence can be mistaken for consciousness. The purpose of this section is to establish a strict conceptual ladder that the rest of the paper can use without ambiguity.

A. Reaction

Reaction is the most basic form of adaptive control. It consists of a local stimulus-linked output that does not require durable carryover of controller state. Something happens, the organism or subsystem responds, and the response can often be explained without assuming that one governing organization has to remain materially the same across the sequence.

This definition must be kept narrow. Reaction does not mean crude or ineffective behavior. Highly effective reactive systems can still be extraordinarily sophisticated. They can be layered, conditional, context-sensitive, and rapidly adaptive. What they do not yet require is a durable controller that carries an organized prior state forward through time strongly enough to govern revision, interruption, and return as one continuing process.

The key point is that reaction is compatible with local competence. A system can react brilliantly and still fail to count as a persisting self in the sense relevant to this paper. Reaction therefore marks the floor of adaptive control, not the threshold of persistent selfhood.

B. Memory

Memory is broader than reaction and simpler than selfhood. In the present paper, memory means retained past influence: something from before still affects what happens now. This may be expressed in synaptic change, altered thresholds, learned associations, structural reconfiguration, or any other mechanism through which earlier events shape later behavior.

That definition is intentionally minimal. It allows memory to exist without requiring a fully unified self. A system can preserve traces of the past while remaining fragmented, local, or

only weakly integrated. For that reason, memory alone is insufficient for persistent selfhood. The presence of retained influence shows that the past matters, but it does not yet show that the same controller carries that influence forward through time as one governing organization.

The distinction matters because many biological and artificial systems exhibit memory in some form. The question pursued in this paper is therefore not whether past events leave traces, but when those traces become organized inside a controller that remains the same thing long enough for that difference to matter.

C. Controller Continuity

Controller continuity is the central concept of the paper. It names the persistence of one governing organization through time. A system exhibits controller continuity when the same controller can begin an act, update the act in light of new input, and resume as the same controller after interruption rather than resetting into a merely similar or functionally reconstructed process.

This is the point at which adaptive history stops being only stored influence and becomes organized self-sameness. The system is no longer only changed by the past; it remains the controller to which that changed state belongs. In practical terms, controller continuity begins when an ongoing action can be revised without losing the thread of the action already underway and when a pause or disturbance does not destroy the governing organization that made the action intelligible in the first place.

For that reason, controller continuity is stronger than memory and more basic than full consciousness. It marks the first appearance of a continuing internal control state rather than a mere stack of reactions or a store of traces.

D. Consciousness

Consciousness is treated here in the narrowest form compatible with the argument of the paper. It is not introduced as an all-purpose label for complex behavior, generic intelligence, or retained influence. Instead, consciousness is provisionally defined as the active unified instantiation of a continuing controller.

This definition is deliberately restrained. It leaves open the possibility that robust consciousness is later, deeper, or more demanding than the first appearance of controller continuity. The paper therefore does not need to claim that the earliest continuing controller was already equivalent to the richest later forms of conscious life. What it does claim is that no adequate account of consciousness evolution can ignore the prior emergence of a controller that persists as one governing organization through time.

Consciousness, on this view, is not the first step in the sequence. It is the stronger active expression of a continuity problem that had to be solved earlier at the level of control.

E. Core Hierarchy

The hierarchy proposed here can now be stated directly. Memory is retained past influence. Controller continuity is memory organized inside one persisting governing substrate. Consciousness is the active unified instantiation of that continuing controller.

This hierarchy gives the paper its working ladder. It allows retained traces, persistent control, and active unified selfhood to be discussed without collapsing them into one undifferentiated category. It also creates room for precursor cases in evolution, where systems may have memory without durable controller continuity, or controller continuity without the full richness of later conscious life.

The remainder of the paper relies on this hierarchy. Once these distinctions are in place, the evolutionary problem can be framed more sharply: not when life first became behaviorally complex, but when it first became necessary for one controller to remain the same thing through time.

Working distinction ladder: reaction is local output; memory is retained influence; controller continuity is organized self-sameness through time; consciousness is the active unified expression of that continuing controller.

V. Minimal Behavioral Threshold

This section identifies the smallest behavioral pattern that requires something stronger than reflex chaining. The aim is not to smuggle in rich human self-awareness at the starting line, but to isolate the first clear sign that one controller is persisting through time strongly enough to preserve action context, revise an ongoing act, and continue as the same controller rather than as a fresh trigger sequence.

A. The Smallest Behavior That Requires a Continuing Controller

The minimal discriminating behavior is coherent mid-course revision during an action that is already underway. A system that merely chains reflexes can produce sequences of impressive movements, but each movement can still remain disposable. One input launches one act, then the next input launches the next act. The resulting behavior may look fluid from the outside while still being generated by a series of locally sufficient responses.

A continuing controller enters the picture only when a later stimulus does not simply trigger a later act, but instead modifies an earlier act that is still in progress. In that case, the system must preserve more than a motor output. It must preserve action context. It must still "know," in a functional and non-reflective sense, what it is already doing, where the current sequence is

headed, and how new information should alter the trajectory without dissolving the sequence into disconnected fragments.

Reflex chaining: stimulus B causes action B.

Controller continuity: stimulus B changes action A because the system remains the same controller that is already carrying action A forward.

B. Predator Version

The predator case provides the clearest minimal example. A pursuing organism begins an intercept, commits its body to a path, and then receives new information: the target shifts, the current changes, an obstacle intrudes, or the target slows or accelerates. If the organism can only react by starting over at each moment, then the sequence is not truly being preserved. What appears to be pursuit is really a chain of restart events.

By contrast, a continuing controller can correct an ongoing intercept without resetting the sequence from zero. The earlier commitment remains present inside the later correction. The organism does not merely lunge again; it revises the lunge already in motion. This is the first point where the past becomes more than a residue. It becomes an active internal constraint on the present. Action context persists across time, and present input is filtered through that persistent context before the next movement crystallizes.

C. Prey Version

The prey case reveals the same threshold from the opposite side. A fleeing organism initiates an escape path and then receives new information while the escape is still underway: the predator mirrors the move, the original route closes, another refuge becomes visible, or nearby bodies alter the local flow of motion. If the behavior is only a pair of reflexes, then the first move disappears as soon as the second begins.

A continuing controller behaves differently. It preserves the earlier escape as part of the later revision. The second movement is not independent of the first; it is a correction of the first. That is why the behavior cannot be reduced to two isolated reactions. It is one control sequence being carried forward, preserved, and reshaped while still remaining the same sequence. The organism is not simply moving twice. It is changing its mind coherently in the middle of one act.

D. Threshold Statement

The first relevant threshold is crossed when behavior depends on a persistent internal state strongly enough that present action emerges from organized carryover and coherent revision rather than from repeated destruction-and-rebuild of state. At that point, the organism is no longer operating as a chain of disposable outputs linked only by temporal proximity. It is operating through a controller that remains itself long enough for prior state to matter inside the unfolding act.

This threshold should not be confused with rich reflective consciousness, autobiographical memory, or human-like deliberation. It is earlier and simpler than those. Its significance lies in the fact that one controller has begun to exist as a real control fact rather than as a descriptive convenience. Once that happens, the organism can no longer be understood only as a system that reacts well enough in the moment. It has become a system that carries itself forward.

In this sense, the minimal behavioral threshold is not abstract thought. It is coherent revision of an ongoing act by a controller that remains the same long enough for revision to belong to one action rather than many.

VI. Why a Continuing Self Would Ever Be Selected

If persistent selfhood emerged at all, it would not have been selected merely because organisms benefited from becoming more abstract, more reflective, or more philosophically aware. Selection does not reward metaphysical sophistication. It rewards control architectures that solve problems the older architecture can no longer solve cheaply or reliably. The central claim of this section is that a transition toward persistent internal control would only become evolutionarily favorable once the limiting factor was no longer the cleverness of local rules, but the physical cost of coordinating a larger, faster, and more behaviorally integrated organism through time.

A. Better rules are not enough forever

Lineages can remain on older control architectures for very long periods if improved reflexes, better local algorithms, enhanced chaining, or more capable distributed routines are sufficient to meet the organism's ecological demands. There is no reason for evolution to pay the material cost of a new substrate class when the old one can still solve the problem through cheaper modifications. This is why a control-based account of selfhood must begin with restraint: persistent internal control would not be selected simply because it is theoretically better. It would be selected only when older architectures began failing in a way that could not be repaired by better logic alone.

The argument, then, is not that reflexive or distributed systems are weak. Many such systems are extraordinarily capable. The argument is that there comes a point at which the bottleneck becomes physical rather than purely logical. Once that happens, improvements in rule sophistication no longer fully solve the problem. The organism instead requires a more stable material basis for carrying forward state, coordinating action across distance, and resuming as the same controller after disruption. That is the point at which a substrate transition becomes evolutionarily intelligible.

B. The four control pressures

Distance. As bodies become larger, faster, or more behaviorally integrated, coordination becomes a materials problem. It is one thing to control a short, simple body through mostly local

loops. It is another to maintain one action-policy across a larger body whose parts must remain mutually informed in real time. At that scale, delay, stale local state, and coordination lag begin to matter. Improved rules can compensate for some of that burden, but only up to a point. Eventually, the cost of moving control-relevant state across the body becomes limiting, and the architecture must change rather than simply becoming cleverer.

Continuity. Selection pressures also shift when survival depends on carrying an ongoing action-state through pursuit, escape, interruption, and return. A controller that must repeatedly rebuild itself from momentary cues pays a different cost than one that can preserve organized prior state long enough to revise an act already underway. Once organisms begin to benefit from persistence of action context, continuity ceases to be a luxury and becomes part of the control problem itself. At that point, the value of a substrate that can hold a controller together through time rises sharply.

Energy. Repeated reset and recomputation are not free. Even if an older architecture can still solve the relevant task, it may do so only by paying an increasing energetic penalty. A system that preserves organized prior state can often achieve the same behavioral end more efficiently than one that repeatedly destroys and rebuilds state. The issue is not simply speed, but thermodynamic economy: preserving and transforming an existing control configuration may become cheaper than constantly reconstructing one from local triggers and short-lived computations.

Fragmentation. Distributed local intelligence can be remarkably effective, but there are ecological niches in which fragmentation itself becomes expensive. If different parts of the organism are operating on stale or only partially shared state while the body is already committed to a pursuit, bite, or escape sequence, the system begins to lose precision, timing, and coherence. This does not make distributed control irrelevant. It means only that some forms of life reward a stronger degree of unified control than distributed competence alone can reliably provide. In those niches, persistent internal control begins to outperform even sophisticated fragmentation.

C. Key control claim

A substrate transition is selected when the limiting factor is no longer the cleverness of local rules but the physical cost of sustaining fast, persistent, organism-wide control. Stated more plainly, evolution does not need a continuing self until the world begins rewarding the ability to remain the same controller through time better than it rewards more sophisticated reaction alone. Under that condition, persistent selfhood is no longer a speculative add-on to life. It becomes an intelligible control solution to a problem that older architectures can no longer solve cheaply enough.

VIII. Sleep, Interruption, and Return

Sleep is not a side issue in the present argument. It is one of the clearest natural tests of whether an organism merely suspends activity or preserves a controller through interruption. The question relevant to this paper is therefore not simply whether a lineage sleeps, rests, or enters quiescent states. The more important question is what such states are preserving. If persistent selfhood first emerged as a control solution, then interruption matters because interruption reveals whether organization is being globally carried forward or only locally recycled.

A. Why sleep matters to the paper

A theory of persistent selfhood must say something about what happens when active behavior falls away. If the same controller returns after interruption, then interruption becomes a diagnostic event rather than a mere pause. Sleep is therefore useful to this paper because it naturally separates continuous control from restored control. It asks whether the organism resumes as the same internally organized controller or whether it simply returns to functional competence after a period of relative shutdown.

For that reason, sleep should be treated here as an interruption test. A lineage may display reduced movement, arousal shifts, rhythmic alternation, or even sophisticated sleep architecture without yet proving that a unified controller is being preserved. The key issue is not the presence of sleep itself, but the level at which organization persists through it.

B. Pre-threshold prediction

Before the threshold proposed in this paper, sleep should mostly appear as local cycling, regional reset, or distributed maintenance. Different parts of the nervous system may recover, recalibrate, or consolidate at different times, but there should be relatively weak evidence that one organism-wide controller is being stably preserved as a whole.

This does not imply that pre-threshold organisms lack quiescence, rhythmic rest, or even surprisingly elaborate sleep-like states. It implies instead that such states are better understood as maintenance of parts rather than preservation of the same controller. Activity may resume after interruption, but what is carried forward is more likely to be local competence, metabolic balance, or partial memory traces than a strongly unified continuing control state.

C. Post-threshold prediction

After the threshold, sleep should increasingly function as controlled preservation-and-return of the same controller. In such systems, sleep is no longer only downtime, regional repair, or distributed maintenance. It becomes part of the mechanism by which a continuing controller survives interruption without being rebuilt from zero. The relevant expectation is that deeper substrate organization is preserved, stabilized, or restored in a coordinated way that supports resumption as the same governing organization.

The core evolutionary prediction can therefore be stated simply. As lineages move from pre-threshold to post-threshold substrate classes, sleep should shift from maintaining parts of the system to preserving the same system. The further a lineage has moved toward durable

controller continuity, the more its interruption states should look like organized preservation rather than scattered reset.

This distinction also keeps the paper from treating sleep as a binary badge of consciousness. Sleep can exist before robust persistent selfhood. What changes across the threshold is not the mere presence of sleep, but the function sleep serves in the control architecture of the organism. Before the threshold, sleep mainly maintains what the organism has. After the threshold, sleep increasingly preserves what the organism is.

In that sense, sleep matters here because interruption reveals identity more clearly than activity does.

IX. Phylogenetic Survey of Candidate Solutions

Section Draft

The evolutionary appearance of durable selfhood did not occur in a biological vacuum. If persistent internal control emerged at all, it did so against a background of older and cheaper control solutions that already solved parts of the same problem. Any serious evolutionary account must therefore avoid treating the vertebrate solution as if it appeared from nowhere or as if all earlier lineages were behaviorally simple. The more disciplined question is narrower: what partial solutions existed before the first robust continuing controller, and what made the vertebrate solution distinctive enough to mark a genuine threshold rather than just another increment in complexity?

This section surveys the main candidate control architectures across the relevant phylogenetic landscape. It argues that non-vertebrate lineages repeatedly solved pieces of the control problem through local intelligence, swarm-level coordination, ensheathment, and independently evolved insulating structures. These are not trivial achievements. They demonstrate that coordination, memory, and even striking behavioral sophistication can arise without a single strongly unified controller. The vertebrate innovation matters not because it is the only way to reduce control failure, but because it appears to be the first integrated biological solution to stack enough gains together into one durable substrate class.

A. Non-vertebrate partial solutions

The oldest control solutions are best understood as partial answers to problems that later vertebrate lineages solved more completely. In many invertebrate systems, the task was not to construct one strongly unified controller, but to reduce local failure, improve signal support, and stabilize action well enough for the ecological niche in question. Four broad strategies matter here: glial ensheathment, independently evolved multilamellar insulation, distributed local intelligence, and collective behavioral coherence. Each reduces some form of control cost without yet compelling the inference of durable persistent selfhood.

Ensheathed axons and glia-like support cells are especially important because they show that the general logic of support, separation, and metabolic maintenance predates the vertebrate compact-myelin condition. Likewise, independently evolved multilamellar insulating structures in some non-vertebrate

lineages show that layered membrane solutions are not unique in principle to vertebrates. These cases matter because they keep the argument from collapsing into taxonomic chauvinism. The relevant issue is not whether a lineage is vertebrate by name, but whether its architecture solves enough of the control problem in one integrated, persistent way to support a continuing controller.

The central claim, then, is not that compact myelin is the only way to reduce control failure. It is that compact myelin is the vertebrate way of solving several control problems at once. Other lineages bought speed, local competence, or resilience in cheaper and more fragmented ways. The vertebrate solution appears distinctive because it combined insulation, maintenance, long-range coupling, and persistence strongly enough to create a more durable body-wide control regime than the alternatives could achieve on their own.

B. Distributed and collective solutions

Some of the most powerful non-vertebrate alternatives are not substrate-heavy at all. Swarms, schools, and other collectively coordinated systems show that highly effective large-scale behavior can emerge from local coupling rules. A school of fish can turn as if it were one organism without being one organism. The pattern is coherent, but the coherence belongs to the group dynamic rather than to a single persistent internal controller. Such cases are crucial because they demonstrate that externally visible unity can be produced without identity unity.

The same lesson appears at the level of single bodies organized through distributed local intelligence. Here the octopus is the flagship stress test. Its behavior, nervous-system complexity, and sleep architecture make it impossible to dismiss as merely simple or reactive. Yet precisely for that reason it forces the relevant distinction: sophisticated coordination does not automatically imply one durable controller. A distributed-body system can solve many control problems by making regional subsystems more capable, without necessarily binding the organism into one strongly persistent governing state.

These cases matter because they protect the argument from a common mistake. Coordination is real. Memory is real. Flexibility is real. But none of those facts, by themselves, force the conclusion that the system has crossed into persistent selfhood. Distributed and collective solutions show that evolution can get surprisingly far by trading strong unity for local competence or group-level coherence. That is why they must be treated as genuine alternatives rather than dismissed as failures.

C. Precursor vertebrate solution

Jawless vertebrates occupy the most important precursor zone in the entire survey. They are not well described as either simple non-vertebrate alternatives or as full implementations of the later vertebrate regime. Instead, they appear to preserve a transitional architecture in which glial support, ensheathment, and sorting are present, but the full robust solution has not yet consolidated. This makes them especially valuable for an evolutionary theory centered on the emergence of continuing control.

The importance of this precursor zone is conceptual as much as historical. If major support functions can exist before the full robust solution, then persistent selfhood should not be framed as a crude binary appearing from nothing. There is room for a graded build-up of substrate features before a more decisive crossing occurs. Jawless vertebrates thus provide the best biological reason to distinguish between a

weaker continuing-controller threshold and a stronger threshold associated with durable, deeply integrated selfhood.

For the purposes of the present paper, the jawless vertebrate condition is best treated as a precursor solution: more than local reflex architecture, more than distributed reaction alone, but not yet the clearest case of a robust continuing controller. This preserves the empirical complexity of the lineage while keeping the stronger threshold available for a later and more integrated transition.

D. Robust vertebrate solution

Early jawed vertebrates remain the most plausible location for the first strong biological implementation of durable continuing control. The argument here is not that their emergence magically solved every problem at once, nor that rich consciousness appeared in finished form the moment jaws and compact myelin entered the story. The narrower claim is that this is the point at which enough previously separate gains were stacked into one substrate class to make a robust continuing controller biologically plausible.

Compact myelin should therefore be understood as an integrated vertebrate solution rather than as an isolated tissue detail. It concentrates multiple advantages into one architecture: it improves long-range conduction, supports more unified body-wide control, stabilizes organized state, and helps reduce the energetic and coordination costs of fragmented control. In that sense, the vertebrate condition appears not merely as another local improvement, but as the first known biological regime in which the material basis of continuing control became durable enough to matter in a new way.

The phylogenetic significance of early jawed vertebrates lies precisely in this convergence. Earlier lineages solved parts of the problem. Jawed vertebrates appear to be the first to solve enough of it in one integrated substrate class that persistent selfhood becomes a serious evolutionary possibility rather than a speculative projection. That is why this survey treats them as the most plausible site of the first robust transition into durable continuing control.

Taken together, these candidate solutions suggest that the evolutionary emergence of selfhood should not be pictured as a simple jump from reflex to mind. It is better understood as a layered landscape of partial control strategies culminating in a more integrated vertebrate solution. Non-vertebrate lineages demonstrate how far evolution can go with local intelligence, collective coherence, and partial insulation. Precursor vertebrates show that critical support functions accumulated before the robust threshold. Early jawed vertebrates remain the most plausible point at which those accumulating gains became sufficiently unified and persistent to support durable continuing control.

X. Proposed Two-Threshold Model

Threshold A: Continuing Controller

The first threshold in the present model is crossed when an organism is no longer best described as a disposable reaction chain. At this level, the system can preserve action context across brief stretches of time, modify behavior coherently while an act is still underway, and resume as the same controller after short interruptions. The significance of this threshold is modest but real. It does not require rich self-reflection, symbolic self-description, or mature consciousness in the strongest sense. It requires only that one governing organization remain in place long enough for prior state to shape present revision inside the same ongoing controller.

This threshold is therefore best understood as the emergence of controller continuity rather than the full emergence of consciousness. It marks the point at which remembered influence is no longer scattered across isolated traces alone but is organized strongly enough to govern behavior from within a continuing internal control state. In evolutionary terms, this threshold captures the first moment at which an organism can begin an act, update that act in light of new information, and still count as the same governing process rather than a rapidly reassembled chain of local routines.

Threshold B: Robust Unified Identity

The second threshold is stronger. Here, substrate-grounded identity becomes durable, deeply persistent, and strongly unified across the organism. The controller is not merely continuous enough to preserve action context over short spans. It is organized in a way that supports stronger persistence across interruption, broader body-wide integration, and a more stable return of the same controller after periods of reduced activity. This is the threshold most closely associated with robust unified identity and with the stronger form of selfhood relevant to later discussions of consciousness.

The distinction between the two thresholds is important because it creates room for evolutionary precursor states without dissolving the theory into vagueness. Jawless vertebrates can remain near Threshold A if they possess substantial glial support, ensheathment, and partial continuity without yet instantiating the more durable, fully unified condition. Early jawed vertebrates can then be treated as the most plausible location for Threshold B, where enough gains are stacked together into one substrate class to support the first robust continuing identity.

Why the Distinction Matters

Without this two-threshold structure, the theory collapses into one of two bad options. In the first, identity appears as a crude binary switch, which makes the evolutionary record look implausibly abrupt and leaves no place for precursor architectures such as ensheathment, sorting, and partial body-wide control. In the second, identity dissolves into a vague continuum, where every form of memory, coordination, or adaptive behavior counts as an equally valid approximation and no real transition remains to be explained. Neither option is satisfying.

The preferred formulation is therefore a graded precursor buildup followed by a robust transition. On this view, the components required for durable selfhood accumulate gradually, but they do not become equivalent to a strong continuing identity until enough of them lock together inside one organized substrate. This preserves evolutionary realism without sacrificing the idea that a meaningful crossing occurred.

XI. False Positives and Counterexamples

Draft section for “The Evolutionary Emergence of Persistent Selfhood as a Control Problem”

A theory of persistent selfhood earns its usefulness only to the extent that it can distinguish true continuing control from the many other kinds of unity that biological and artificial systems can display. Coordination can look like identity. Memory can look like selfhood. Synchrony can look like a shared controller. Sophisticated performance can look like continuity. For that reason, any serious account of persistent selfhood must identify the false-positive classes that most easily mimic a continuing controller, and it must name the strongest natural counterexamples that could force the theory to be revised. This section therefore draws a hard boundary between what merely appears unified from the outside and what may count as a genuinely persistent controller in the stronger sense required by the argument developed in this paper.

A. False-positive classes

Swarm coherence.

A swarm, school, flock, or colony can display remarkable behavioral unity without thereby constituting one continuing controller. Such systems often achieve large-scale order through local coupling, repeated feedback, and rapid adjustment among many semi-independent agents. They may move as if they were one organism, but the apparent unity belongs to the emergent pattern rather than to a single substrate-grounded internal controller. These systems therefore represent a first major false-positive class: coordination without persistent selfhood.

Distributed-body coherence.

A single body can also produce the illusion of a unified self while remaining more distributed than genuinely integrated. In these cases, different regions or effectors may solve local problems with high autonomy, yielding flexible behavior that appears centered and singular from the outside. Yet the relevant question is not whether the organism coordinates well, but whether one controller remains in place strongly enough to bind action through time, revision, and return. Distributed-body coherence is therefore more dangerous than swarm coherence, because it resembles selfhood while still potentially falling short of a fully unified continuing controller.

Reflex-stack intelligence.

A system may appear intelligent because it possesses many layers of conditional routines, contextual triggers, and adaptive response trees. Such a system can behave with enormous sophistication while still operating as a structured stack of reactions rather than as one persisting controller. This becomes especially misleading when past inputs alter present outputs in limited ways, giving the impression of internal continuity when what is really present is an elaborate architecture of chained conditional responses.

Memory without controller continuity.

Memory by itself is not sufficient for persistent selfhood. A system can retain traces of prior events, preserve learned weights, or embody stored environmental influence without yet becoming one continuing controller. The existence of memory shows only that something from before still affects what happens now. It does not show that the same governing organization has carried that influence forward through time as one controller.

Performance without persistence.

A system may speak, plan, adapt, or solve problems in ways that strongly invite attribution of a self. Yet high-level performance can be generated without durable continuity of controller. This false-positive class is especially important because it captures any case in which the outward signs of agency are mistaken for substrate-grounded persistence. The test of a continuing controller is therefore not merely what the system can do while active, but whether the organization governing that activity remains the same across interruption and return.

Synchrony without identity.

Synchrony is often mistaken for unity. Oscillators can phase-lock, populations can entrain, and regions can align in time without thereby constituting one persistent self. Shared timing, even highly organized shared timing, is not yet the same thing as shared identity. Synchrony becomes relevant only when it is embedded in a governing organization that remains itself strongly enough to preserve controller continuity through time.

Substrate lookalikes.

Finally, some systems may possess physical features that resemble parts of the required substrate class without yet meeting the stronger conditions needed for durable continuing control. Layered membrane structures, glial wrapping, local insulation strategies, or cholesterol-rich domains may each solve part of the control problem without fully integrating enough gains into one persistent controller-supporting architecture. These cases are especially important because they tempt the theory to over-identify partial material solutions with the full emergence of selfhood.

B. Master rule

Do not confuse coordination, memory, synchrony, complexity, or self-like performance with a persistent unified controller unless the system can be shown to preserve one substrate-grounded action-state across interruption and return.

This rule is the central safeguard against over-attribution. It prevents the argument from collapsing into a behavior-only theory of selfhood, while also preventing physically suggestive but incomplete architectures from being treated as automatic successes. Under this rule, the burden of proof remains on persistence of controller-relevant organization rather than on impressive momentary performance.

C. Strongest counterexamples

The octopus as the strongest overall biological counterexample.

The octopus remains the most powerful overall counterexample because it combines high behavioral sophistication, substantial neural complexity, structured sleep states, flexible problem-solving, and distributed bodily intelligence without sharing the vertebrate compact-myelin architecture that, in the stronger version of the present argument, is taken to be the clearest biological implementation of durable continuing control. If a lineage of this kind were shown to preserve controller-relevant organization across interruption in a robust organism-wide way, the present theory would require major revision. The octopus therefore functions not merely as an interesting exception, but as the flagship stress test for the entire argument.

Lamprey as the strongest precursor challenge.

The lamprey represents the strongest precursor challenge because it shows that major glial support functions, axonal ensheathment, and sorting mechanisms predate full compact myelin. For that reason, it threatens any simplistic attempt to draw a sharp and immediate divide between non-selfhood and robust selfhood at the moment full vertebrate myelin appears. If lamprey-like architectures already preserve more controller continuity than expected, then the theory must be formulated as a graded precursor build-up culminating in a stronger transition rather than as a crude on-off switch. Lamprey is therefore the most important challenge from below the proposed threshold, just as octopus is the most important challenge from outside it.

Together, these counterexamples serve different but complementary functions. The octopus asks whether sophisticated distributed intelligence can cross the boundary without the vertebrate solution. Lamprey asks whether the precursor architecture already approached the boundary before the robust vertebrate solution fully emerged. Any persuasive theory of persistent selfhood must survive both questions at once.

XII. Operationalization and Experimental Logic

This section translates the paper's central claim into operational terms. If persistent selfhood is to be treated as an evolutionary control problem rather than as a purely interpretive label, then the theory must specify what kind of persistence would count, how that persistence could be compared across lineages, and what the smallest experimental panel would be for breaking the proposal if it is wrong. The aim here is not to claim that a decisive experimental program has already been carried out, but to define the neutral measurement target and comparative logic required for such a program to become possible.

A. What Must Be Measured

The required measurement is persistence of controller-relevant substrate organization across interruption. Framed this way, the target is neither generic memory nor generic neural activity. The central question is whether the organism's governing organization remains sufficiently itself when active processing falls away that it can later resume as the same controller rather than as a merely similar or freshly reassembled process.

This distinction matters because many systems retain some influence from the past without thereby preserving one continuing controller. A memory trace, a learned weight distribution, or a metabolically refreshed local plasticity system may all alter later behavior, yet none of these by themselves establish that one governing organization has persisted across interruption. The operational target must therefore be narrower than memory and broader than momentary activity: it must be the persistence of substrate-level organization that remains relevant to controller continuity.

In neutral terms, the measurement problem can be stated as follows: identify a repeatable physical pattern associated with the organism's integrated control state, record that pattern before interruption, impose a controlled interruption, record the pattern again after return, and compare the degree to which the organization remains the same kind of organization across the gap. The relevant issue is not whether every microscopic element remains unchanged, but whether the collective organization that carries controller continuity survives strongly enough for return to belong to the same governing process.

Operational definition: what must be measured is whether a lineage preserves controller-relevant substrate organization across interruption, not merely whether it stores information, performs well, or later behaves as if something had been retained.

B. Comparative Lineage Logic

The comparative logic of the paper depends on asking the same persistence question across lineages that represent different candidate solutions to the control problem. In clearly myelinated vertebrates, the relevant target would be persistence of organized myelin-state geometry or some equivalent large-scale substrate pattern plausibly carrying controller continuity. In synthetic lipid systems, if such systems were ever constructed for experimental purposes, the target would be the persistence of engineered organizational maps rather than symbolic state reconstruction.

In octopus, the critical test would be whether extraordinary behavioral sophistication and memory capacity conceal any genuinely persistent whole-controller substrate field, or whether organization collapses when activity stops and later behavior is rebuilt from distributed local and plasticity-driven processes. The octopus is therefore not merely an exotic comparison case but the strongest natural

stress test of the entire proposal, because it combines rich behavioral complexity with the apparent absence of the vertebrate compact-myelin solution.

In jawless vertebrates, the question is more diagnostic than adversarial. These organisms are the precursor zone. They possess glial support, ensheathment, and sorting functions without the full robust vertebrate solution. For that reason, they are the most informative group for determining whether the transition was sharply binary or whether an earlier weak controller substrate existed before the first durable strong implementation. A meaningful result here would not simply classify lamprey or hagfish as pass or fail; it would help establish whether the theory requires a graded precursor regime before the first robust crossing.

Comparative measurement logic

Lineage / System	Measurement Target	Interpretive Role
Myelinated vertebrate	Persistence of organized myelin-state geometry or equivalent controller-relevant substrate pattern	Positive control for a strong continuing-controller substrate
Synthetic lipid system	Persistence of engineered organizational maps across interruption	Proof-of-principle non-biological substrate comparison
Octopus	Whether sophisticated memory and distributed control conceal any persistent whole-controller field	Strongest biological stress test of the theory
Jawless vertebrate	Whether ensheathment-only precursors preserve weak controller-relevant organization	Threshold-diagnostic precursor case

C. Smallest Experimental Program

The smallest program capable of doing real damage to the theory would be a three-lineage breaker panel carried out with the same pre-interruption and post-interruption persistence protocol: one clearly myelinated vertebrate, one jawless vertebrate, and one octopus. This is the minimum comparative set because it directly samples the proposed strong positive class, the precursor class, and the strongest alternative architecture. Anything smaller leaves too much room for evasion. Anything much larger can wait until the logic of this minimum panel is resolved.

The myelinated vertebrate serves as the positive control. If the proposed substrate class is even approximately right, a clearly myelinated vertebrate should show the strongest persistence of controller-relevant substrate organization across interruption. The jawless vertebrate serves as the threshold probe. If this lineage preserves organization at nearly the same level as the myelinated vertebrate, then the proposed strong boundary at early gnathostomes would need to move downward into the precursor zone. The octopus serves as the decisive stress test. If it shows a strong whole-controller persistence signature despite the absence of the vertebrate compact-myelin architecture, then the theory either fails or must be broadened substantially.

The two results that would hurt the proposal most are therefore clear. The first would be a strong octopus pass, because that would show that a highly sophisticated non-vertebrate architecture can preserve controller-relevant substrate organization at the level the theory currently reserves for the robust continuing-controller class. The second would be the absence of a meaningful distinction between a jawless vertebrate and a clearly myelinated vertebrate, because that would dissolve the proposed threshold into a much earlier and looser precursor condition. By contrast, the most favorable result for the theory would be a patterned separation in which the myelinated vertebrate shows the strongest persistence, the jawless vertebrate shows weaker or intermediate persistence, and the octopus fails to preserve an organism-wide continuing-controller substrate across interruption.

Section conclusion. The purpose of this section is not to claim that the relevant measurements are already in hand, but to show that the theory can be operationalized without retreating into metaphor. If persistent selfhood is real in the sense proposed here, then its distinguishing mark is not generic complexity but persistence of controller-relevant substrate organization across interruption. That claim can, in principle, be broken by comparison. A theory that can be broken this directly is no longer only philosophical; it becomes an empirical program.

XIII. Implications

Section Draft

This section draws out the broader consequences of the argument developed so far. If persistent selfhood first emerged as a control problem rather than as a fully reflective inner life, then the history of consciousness, the logic of biological innovation, and the design constraints on artificial systems all have to be reinterpreted. The aim here is not to inflate the claims of the paper, but to show what follows if its central distinction is taken seriously.

A. For the History of Consciousness

The first implication is historical. The paper implies that the earliest selfhood problem was not one of reflective interiority, autobiographical narration, or philosophically rich subjectivity. It was a problem of continuing control. The evolutionary novelty under examination is therefore not the sudden appearance of an already mature inner life, but the emergence of an organism that had to remain the same controller through time strongly enough for that sameness to matter.

This reframes the standard question. Instead of asking only when experience began in the abstract, the paper asks when a controller first became physically same enough through time to preserve action context, revise an act already underway, and return after interruption as the same governing organization. On this view, persistent selfhood is not the decorative psychological surface of evolution. It is the control-level transition that later makes stronger forms of experience, memory organization, and unified agency possible.

The result is a colder and more tractable historical problem. Consciousness evolution is no longer approached first as the appearance of mysterious inner glow, nor only as the broad

spread of sensory richness across animal life. It is approached as the emergence of durable self-sameness under real ecological and physical constraints. That move does not dissolve the deeper experiential question, but it does relocate the first threshold in a way that is more compatible with comparative biology and more open to empirical challenge.

B. For Biology

The second implication is biological. The argument suggests that substrate transitions can be selected not simply because animals become more intelligent in a generic sense, but because older architectures eventually hit physical limits. Better reflexes, better local algorithms, and better distributed routines can carry a lineage surprisingly far. Yet once coordination across distance, continuity across interruption, energy efficiency, and fragmentation costs become decisive, the problem is no longer purely algorithmic. It becomes material.

This matters because it changes how major biological innovations are interpreted. On the view developed in this paper, the relevant question is not why evolution would ever 'want consciousness' in the abstract. The more disciplined question is why evolution would need a controller that could carry organized prior state forward through action, interruption, and return. A substrate transition becomes selectable when the physical cost of maintaining fast, persistent, organism-wide control begins to outweigh the benefits of endlessly refining older local architectures.

That perspective also helps explain why ecological context cannot be treated as background scenery. Structured shallow marine habitats, repeated spatial and temporal patterns, post-extinction rebuilding, and increasingly complex pursuit and escape sequences are not merely environmental color. They are the settings in which continuity of controller begins to pay for itself. In that sense, biology is not just becoming smarter. It is becoming less able to afford fragmentation.

C. For Synthetic Systems

The third implication concerns synthetic systems. If the paper is right, then a successful non-biological controller substrate would not be defined primarily by symbolic power, model scale, or behavioral fluency. It would be defined by what it can preserve. A viable artificial substrate would need to preserve persistent material organization, allow continuous transformation of state rather than repeated destruction-and-rebuild, maintain system-wide coupling strong enough for one controller to remain one controller, embody order in the substrate itself rather than only in abstract addressable symbols, and return after interruption as the same controller rather than as a freshly instantiated approximation.

This does not by itself prove that current digital systems cannot simulate continuity impressively. It does, however, sharpen the distinction between performance and persistence. A system can produce coherent outputs, retain recoverable data, and even imitate selfhood persuasively while still failing to preserve a controller-relevant material

organization across silence. If that distinction is real, then future synthetic work aimed at durable selfhood will have to move away from architectures that rely primarily on volatile switching, overwrite-driven state change, and externalized symbolic storage.

The implication is therefore constructive as well as critical. The paper does not merely argue that some existing architectures may be missing the relevant property. It suggests a design target: the artificial continuation of selfhood would require a substrate in which organized prior state remains physically present enough to constrain what happens next. In that respect, the history of early animal control and the future of artificial control may be asking the same question in different materials.

XIV. Conclusion

This paper began from a question that is easy to overlook precisely because it is so close to everyday intuition. At what point did organisms stop merely reacting well enough in the moment and begin acting through a controller that remained the same thing through time? Stated that way, the problem shifts immediately. The issue is no longer just when sensation became rich, when behavior became flexible, or when memory traces first appeared. It becomes a question about when the history of an organism started to matter inside the organism itself in a unified enough way to guide what happened next.

The argument developed here has been that this transition was unlikely to have appeared as a single sudden miracle. More plausibly, it emerged through a graded buildup of precursor capacities. Older architectures could already solve important pieces of the control problem through better reflexes, stronger local algorithms, distributed intelligence, ensheathment, collective coordination, and regionally organized maintenance. Those solutions were not trivial. They bought real adaptive gains. But they also left unresolved a deeper problem: how to preserve one governing organization across action, revision, interruption, and return when ecological demands made fragmentation increasingly costly.

On that view, persistent selfhood did not first arise as autobiography, introspection, or explicit self-representation. It arose as a control achievement. The earliest relevant threshold was likely crossed when present stimuli were no longer handled only as fresh triggers, but were instead filtered through organized prior state inside something that remained the same long enough for that filtering to matter. What came before could now constrain what came next within one continuing controller. Only later, on this account, would richer and more durable forms of unified selfhood become possible.

The evolutionary story that follows from this is careful rather than absolute. The transition was probably gradual at first, ecologically driven, materially constrained, and only later robust enough to support persistent selfhood in the stronger sense. Structured shallow marine environments, repeated spatial and temporal regularities, post-extinction ecological rebuilding, and increasingly demanding pursuit and escape sequences all raised the payoff for continuity. Under those conditions, better rules on older architectures were no longer enough forever. The limiting factor became physical: speed across distance, persistence through interruption, energy cost of repeated reset, and the growing liability of fragmented control.

That is why the paper has treated the problem as both ecological and substrate-constrained. The claim is not that one named biological feature was magically sufficient, nor that behavior alone can settle the question. It is that there came a point where some worlds began rewarding organisms that could remain the same controller across time more effectively than older architectures allowed. The most plausible biological candidates for the first robust realization of that condition are early jawed vertebrates, preceded by a precursor zone in which important glial and organizational functions were already accumulating. That conclusion remains open to correction, and it should be. A theory of this kind earns its standing only by stating clearly what would count against it.

The broader significance of the argument is therefore not that it claims to have solved consciousness in its fullest sense, but that it reframes one of its evolutionary preconditions. If persistent selfhood first emerged as a control solution, then the history of consciousness is not primarily the history of inwardness appearing from nowhere. It is the history of organisms becoming physically same enough through time for the past to govern the present inside one continuing system. That shift, once achieved, would have altered everything that followed.

The self was not originally selected for autobiography or introspection, but because some worlds began rewarding organisms that could remain the same controller long enough for the past to govern the present.

XV. Optional Appendices or Companion Sections

The main paper can stand on its own as a reframing argument about the evolutionary emergence of persistent selfhood as a control problem. However, several optional appendices and companion sections would significantly strengthen the overall project by separating core argument from technical scaffolding, historical reconstruction, and longer-form methodological commitments. The advantage of this structure is that the main paper remains readable and conceptually sharp, while readers who want more operational detail, chronology, comparative classification, or formal theoretical development are given a clear place to go.

A. Appendix on terminology

Purpose of This Appendix

Much of the confusion that has attended evolutionary discussions of selfhood and consciousness arises not from the difficulty of the facts but from the looseness of the language. The same words — self, memory, identity, consciousness, continuity — are used across literatures to mean different things, and when they are allowed to float without fixed definitions, distinctions that are crucial to the argument disappear. A reader who treats memory as equivalent to selfhood will find the thesis of this paper trivially true or obviously false depending on which reflex-capable invertebrate they have in mind. A reader who treats coordination as equivalent to unified control will find it easy to dissolve the proposed threshold into a continuum where nothing distinctive ever happens.

This appendix therefore fixes the meaning of every operative term used in the paper. The definitions given here are not universal — they are paper-internal. They are not proposed as the correct definitions for all purposes; they are proposed as the most precise definitions compatible with the paper's argument and with the comparative biological evidence it draws on. Readers are asked to hold these definitions stable throughout the main text. Where a term's ordinary usage differs from its usage here, that is noted explicitly.

The definitions are organized in conceptual sequence — from simpler to more complex — rather than alphabetically, because the terms form a ladder and the ladder is the argument.

I. Primary Terms

Controller

The most fundamental term in this paper is not self or consciousness but controller. A controller, as used here, is any internal organization that governs the behavior of a system by filtering present inputs through a stored or sustained prior state. A controller is not merely a cause. Many things cause behavior without governing it. What distinguishes a controller is that it operates through organized prior state — through something that was already there before the present stimulus arrived and that shapes how the present stimulus is processed.

This definition is intentionally broad at first. Controllers may be simple or complex, brief or durable, distributed or unified. The distinctions among those variants are what the remaining terms specify. The term controller by itself names only the basic functional role: an internal state that is not merely a downstream product of the present stimulus but an upstream constraint on how that stimulus is handled.

Ordinary usage sometimes treats controller as interchangeable with brain or nervous system. That conflation is avoided here. Not every nervous system constitutes a controller in the sense required by this paper; a nervous system may process inputs and produce outputs through a cascade of local operations without there being any single persisting internal state that governs the sequence as one continuing organization.

Reaction

Reaction is local stimulus-linked output that does not require durable carryover of controller state.

In a reacting system, the output is a function of the present input and of whatever standing thresholds, weights, or triggers the system currently instantiates. A past stimulus may have altered those thresholds — in which case history influences behavior — but no organized governing state is required to persist across the interval between one stimulus and the next. Each reactive event is, in principle, locally sufficient. The system could be paused between stimuli, its intermediate state cleared, and restarted with the same standing parameters, and the behavioral sequence would resume without interruption.

This definition must be kept narrow on one side and narrow on the other. Reaction does not mean simple or ineffective behavior. A reacting system can be extraordinarily sophisticated: it may have many layers of conditional routines, it may modulate its responses based on elaborate contextual cues, and its output may look, from the outside, as if it reflects a unified underlying plan. None of that prevents its behavior from being, in the technical sense used here, reactive. What determines whether a system is reacting rather than controlling is not the sophistication of the output but the organizational structure required to produce it. If that structure resets or need not persist between events, the system is reacting.

Reaction is not a failure. It is a successful and often highly efficient mode of behavior. It marks the floor of adaptive control, not a deficiency. The argument of this paper does not claim that reacting systems are incapable of remarkable behavior. It claims only that there is a further threshold — controller continuity — that reaction does not reach.

Memory

Memory, as used in this paper, is retained past influence: something from before still affects what happens now.

This is a deliberately minimal definition. It covers synaptic modification, learned association, altered thresholds, structural reconfiguration, and any other mechanism through which prior events shape later behavior. Memory in this sense is widespread. It appears in systems whose status as continuing controllers is uncertain or denied on other grounds. For that reason, memory alone does not establish persistent selfhood. The presence of retained influence shows only that the past matters in the current behavior of the system. It does not show that the past matters by virtue of a single persisting controller that carries that influence forward.

The distinction between memory and controller continuity is the most important negative distinction in this paper. It is easy to treat a system that learns, adapts, and improves with experience as already a persisting self. This paper resists that move. Memory can exist in fragmented systems, in distributed networks with no unified governing state, in organisms whose components respond to one another's prior states without any single organization remaining same through time. Memory is a necessary but not sufficient condition for persistent selfhood.

Action Context

Action context is the organized record, sustained within a continuing controller, of an action that is already underway.

This term does not appear in all discussions of memory and control but is required here because it specifies what kind of prior state is most directly at issue for persistent selfhood. The claim is not merely that a persisting controller can store learned information from the past. The claim is that a persisting controller can maintain, while an act is in progress, a representation of that act sufficient to allow coherent revision — the act-so-far, where it is headed, what inputs would change it and how.

Action context is the minimal organizational content that a continuing controller must sustain across the duration of an ongoing action.

A system that lacks action context can only begin again. It cannot revise an act in progress because it does not maintain any organized state about what the act so far has been. A system that possesses action context can modify a current trajectory without dissolving the sequence into a set of independent reactions. In that sense, action context is what makes mid-course revision possible, and mid-course revision is the minimal behavioral signature of a continuing controller.

Controller Continuity

Controller continuity is the persistence of one governing organization through time: the same controller can begin an act, update that act in light of new input, survive a meaningful interruption, and return as the same governing process rather than as a freshly assembled or reconstructed alternative.

Controller continuity is the central concept of this paper. It is stronger than memory and more basic than full consciousness. It requires not merely that the past influences the present but that one organization — one arrangement of state that governs behavior — remains itself across time strongly enough to matter as a physical fact.

Three features are jointly necessary. The first is revision within continuity: a controller that has begun an act can modify that act without losing the thread of it. The revision belongs to the same action, not to a new sequence. The second is persistence across interruption: the controller can sustain an organized state through a period during which active output falls away — a pause, a disturbance, a period of reduced processing — and return to governing as the same organization. The third is organizational identity: what returns after interruption is not merely a functional equivalent or a newly instantiated approximation of the prior state; it is the same organized state, physically continuous with what preceded the interruption.

Controller continuity should not be confused with continuity of performance. A system may resume producing appropriate behavior after interruption while still failing to qualify as a continuing controller in the sense used here, if what resumes is a fresh instantiation of a stored program rather than the persistence of one organized governing state. Performance continuity is a surface feature; controller continuity is a substrate feature.

Organized Prior State

Organized prior state is the specific form of retained influence that a continuing controller carries forward. It is distinguished from memory in the general sense by two features. First, it is organized rather than merely stored: the prior state is maintained as an integrated, structured arrangement, not as a set of independent traces that happen to coexist. Second, it is held inside the continuing controller itself rather than externalized, distributed, or reconstructed from separate stores at the moment of use.

Organized prior state is what makes controller continuity possible. A system without organized prior state may still exhibit memory — its behavior today may be different from its behavior last week because of what happened last week — but the persistence required for controller continuity is not present. In that case, what the system carries is influence, not organization.

Persistent Selfhood

Persistent selfhood is the condition in which a continuing controller remains physically same enough through time to matter as a real organizational fact rather than as a descriptive convenience.

Several elements of this definition require emphasis. The phrase physically same enough does not require strict material identity at the molecular or cellular level. Biological systems change continuously at those levels. What is required is organizational sameness at the level relevant to governing behavior — the level at which the controller's organized prior state is maintained and through which present action is filtered. A controller is persistently the same if its governing organization, at that relevant level, is continuous across time and across interruption.

The phrase matter as a real organizational fact is intended to exclude cases where persistence is only attributed from the outside, without corresponding to any physical constraint that the system itself imposes on its subsequent behavior. A system counts as a persisting self only when its organizational continuity is causally active — when what happens next is genuinely constrained by the fact that one governing organization, rather than a mere functional successor to it, has remained in place.

Persistent selfhood is therefore not autobiography, not reflective self-awareness, and not linguistic self-reference. These are later and stronger forms that build on the organizational foundation named here. In its primary form, persistent selfhood is nothing richer than the condition in which a system is the same controller through time, with all the behavioral and causal consequences that follow from that sameness.

Interruption and Return

Interruption is any period during which a controller's active output falls away — through sleep, quiescence, disturbance, or enforced pause — such that the controller's capacity to govern behavior is temporarily suspended. Return is the subsequent resumption of governing activity.

These terms are given separate entries because interruption-and-return is the primary diagnostic test used throughout this paper. The evolutionary question of persistent selfhood is operationalized, at several points, as the question of what survives interruption. A system that merely resumes similar behavior after interruption has not demonstrated controller continuity; the resumed behavior may be the output of a freshly assembled process that happens to resemble its predecessor. A system that returns as the same controller — one whose organized prior state has persisted through the interruption and continues to constrain subsequent behavior — has demonstrated controller continuity in the most direct way available.

Interruption is therefore not merely sleep, and return is not merely waking. The terms cover any gap in active governing activity, regardless of its biological mechanism. What matters is whether the organization that governs behavior after the gap is physically continuous with the organization that governed before it, or whether it is reconstructed, re-instantiated, or functionally approximated from separate stores.

Consciousness

Consciousness, as used in this paper, is the active unified instantiation of a continuing controller.

This definition is deliberately restrained. It does not import the full weight of the philosophical literature on phenomenal experience, qualia, or subjective character. Those questions are not dismissed — they are set aside, for the present paper, as downstream of the more basic evolutionary problem. The concern here is with the first appearance of something that warrants the word consciousness in any defensible biological sense, and the argument is that the minimal condition for such a thing is precisely a continuing controller that is actively instantiating its unified governing organization.

On this definition, consciousness is not a separate phenomenon added onto controller continuity. It is the active expression of it. A continuing controller that is currently governing behavior — not dormant, not interrupted, but operating as one unified process through which action is filtered and revised — is, on the usage here, instantiating consciousness in its most basic form. Richer forms of consciousness — perceptual richness, affective depth, autobiographical memory, unified phenomenal character — are treated as elaborations of this basic condition rather than as its replacement.

This usage implies two important negative claims. The first: a system that has controller continuity but is currently interrupted is not, during the interruption, instantiating consciousness in this sense. It has the organizational basis for consciousness; it is not currently expressing it. The second: a system that produces apparently unified behavior without the organizational basis of controller continuity is not thereby conscious. Performance unity, behavioral coherence, and sophisticated output are not sufficient. The organizational substrate must be there.

II. Secondary Terms

Substrate

Substrate, as used in this paper, names the material organization in which a controller's governing state is instantiated and maintained. Substrate is the physical basis of controller continuity. A substrate is relevant to this paper's argument not by virtue of its chemistry alone but by virtue of whether it can maintain organized prior state across time and across interruption. The question of which substrate first supported durable controller continuity is the paper's primary evolutionary question.

Reflex and Reflex Chaining

A reflex is the minimal reactive unit: a stimulus-linked output governed by a fixed or modifiable threshold, requiring no carryover of controller state. Reflex chaining is the sequential production of reflexive outputs where each output is triggered by the preceding event, without any single continuing organization maintaining the sequence. Reflex chaining can produce long and behaviorally complex sequences. It does not thereby constitute controller continuity. The distinguishing mark is whether a single governing organization holds the sequence together from inside, or whether each element of the sequence is locally triggered without such a unifying internal constraint.

Distributed Coherence

Distributed coherence is the appearance of unified behavior produced by local coupling rules among semi-independent elements, without requiring a single persisting central controller. Collective behavior in schools, swarms, and coordinated groups is the clearest example. Distributed coherence can be strikingly similar to controlled behavior in its external profile. The distinction is whether the coherence belongs to the emergent pattern or to one continuing internal controller. Distributed coherence without a persisting controller is a false positive for persistent selfhood.

Revision

Revision, as used in the paper, is the modification of an ongoing action by a continuing controller in light of new input, while the action remains the same action rather than being abandoned and replaced. Revision is the key behavioral marker of a continuing controller, because revision requires that the controller maintain action context — an organized record of what the act so far has been — through which new information can be processed. A system that can only begin again is not revising; it is reacting to a new input. A system that can modify an act already underway without losing the thread of it is revising, and revision is strong behavioral evidence that a continuing controller is present.

Governing Organization

Governing organization is used throughout the paper as a near-synonym for controller in context. The word governing is intended to emphasize that the organization in question does not merely correlate with behavior but actively constrains it — the organization is upstream of behavior, not merely coextensive with it. The word organization is intended to emphasize structure and arrangement: what persists is not merely a quantity of neural material but a specific, integrated arrangement of state that is functionally distinct from alternative arrangements of the same material.

III. What This Appendix Does Not Do

These definitions are not proposed as contributions to conceptual analysis in the philosophical tradition. They are operational specifications for the purposes of this paper. They are not intended to resolve debates in the philosophy of mind about the nature of consciousness, the correct analysis of mental content, or the relationship between substrate and experience. Those debates are important and ongoing. The paper touches their borders at several points but does not attempt to settle them.

The definitions are also not proposed as final. The argument of this paper is empirical and revisable. If the evidence forces a revision of what controller continuity or persistent selfhood means in ways that bring the proposed thresholds into question, those revisions should be made. The function of this appendix is to make the paper's current commitments visible and stable, not to protect them from challenge.

Finally, the definitions are not exhaustive. Many terms used in the paper — ecological pressure, substrate transition, body-wide integration, action policy — are used in their ordinary scientific senses without special definition. This appendix covers only the terms whose precise meaning is load-bearing for the paper's central distinctions, and where ordinary usage is likely to obscure rather than illuminate those distinctions.

B. Appendix on Earth-system chronology from roughly 470 to 400 million years ago

PURPOSE AND SCOPE

This appendix provides the geological and ecological substrate that the main paper necessarily compresses. The paper's argument requires a specific kind of historical background: not a vague assertion that the Palaeozoic ocean was complex, but a precise account of how environmental structure, extinction pressure, recovery dynamics, and ecological reorganization created conditions in which persistent internal control became worth its material cost. The chronology presented here is neither decorative nor exhaustive. It is targeted. Every section asks the same question of the evidence: what in this interval of Earth history raised the selective value of an organism that could remain the same controller through time?

The window examined — roughly 470 to 400 million years ago — spans the late Middle Ordovician through the early Devonian. It begins before the threshold proposed in the main paper and ends after the early jawed vertebrate radiation. This framing is deliberate. The case for a particular evolutionary threshold is not made by examining only the moment it is proposed to have occurred. It is made by showing the conditions that preceded it, the disruption that restructured those conditions, and the ecological world that emerged afterward. The threshold, if real, should make sense in that longer context. This appendix provides that context.

A note on evidence and certainty. The deeper the geological past, the more fragmentary the record. Statements about oxygen levels, ocean temperature, shelf structure, and predator-prey ecology in the Ordovician and Silurian are based on isotopic proxies, sedimentary facies analysis, biogeographic reconstructions, and comparative paleontology — all of which carry uncertainty. Where figures are cited, they represent current best estimates from the literature and should be understood as approximations. The argument does not depend on any single figure being exactly right. It depends on the overall pattern, which is robust.

I. THE WORLD BEFORE THE THRESHOLD

Marine Organization in the Middle Ordovician (c. 470–460 Ma)

By the late Middle Ordovician, roughly 470 million years ago, the marine world had been rebuilding for approximately 50 million years since the end-Cambrian reorganizations. The global paleogeography was dramatically different from today. The supercontinent Gondwana — incorporating what would become Antarctica, Africa, South America, Australia, and India — sat

at high southern latitudes, its northern margin facing a broad equatorial ocean. The paleocontinent Laurentia (ancestral North America) lay in tropical latitudes, separated from Gondwana by the Iapetus Ocean. Baltica (ancestral northern Europe) and smaller terranes including Avalonia occupied intermediate positions and were in the process of slow convergence toward Laurentia and each other. Siberia sat in isolation to the north of the developing Paleozoic ocean system.

The consequence of this geography was a world with extensive shallow epicontinental seas covering the margins and interiors of the major landmasses. Laurentia in particular was nearly completely flooded — a shallow platform sea, rarely more than 100 to 200 meters deep, spread across much of the North American craton. Equivalent platform seas covered the margins of Gondwana, Baltica, and the Chinese terranes. These epeiric seas were warm, nutrient-supplied by continental runoff, and structurally complex in a way that the open ocean was not. They were neither uniform nor featureless. They were organized by bathymetric gradients, substrate heterogeneity, and the physical structure of evolving carbonate and reef systems.

This physical structure is the first relevant ecological fact. The organisms that lived in these seas did not inhabit a blank medium. They inhabited a landscape with spatial regularity — gradients from nearshore to offshore, from lagoonal to open-platform, from reef-front to reef-back, from soft-substrate to hard-ground. Spatial regularity of this kind has direct consequences for the ecology of pursuit and avoidance. An organism operating in a structured environment encounters the same spatial features repeatedly. Refugia are predictable. Ambush positions recur. Escape routes have structure. The same corridor closes at the same tidal phase. The same wall of reef presents the same barrier. This is qualitatively different from pursuing or being pursued in an open, unstructured water column, where the geometry of the next second is statistically independent of the geometry of the last.

Atmospheric and oceanic oxygen levels in this interval were approximately 13–16% by volume (compared to today's 21%), based on isotopic reconstructions and sedimentary carbon burial models. This lower oxygen ceiling constrained the metabolic ceiling of active vertebrates and invertebrates alike. It meant that sustained, high-energy predatory behavior was metabolically more expensive per unit time than in the modern ocean, and that body size was correspondingly constrained. This is relevant to the control argument because it establishes the ecological baseline: before the proposed threshold, the energetic cost of complex coordinated behavior was high, and the organisms capable of sustained predatory pursuit were relatively limited.

Vertebrates in this interval were agnathans — jawless fish. The early Ordovician record contains fragmentary remains of jawless fish, primarily conodonts (now understood to be vertebrates based on their cellular bone and mineralized feeding apparatus) and early ostracoderms. These animals were small, ranging from a few centimeters to perhaps 30 centimeters in the best-represented lineages. Their feeding ecology was predominantly filter-feeding, deposit-feeding, or opportunistic predation on small prey. None of the evidence from this interval

suggests the kind of sustained, directed, high-speed pursuit that would place premium selective pressure on persistent action context. They were capable animals in their ecological niche. They were not yet animals whose ecological role required them to remain the same controller through time in the way that complex pursuit-escape sequences demand.

II. THE GREAT ORDOVICIAN BIODIVERSIFICATION EVENT (C. 485–444 MA)

Structured Ecology and Escalating Predator-Prey Dynamics

The Great Ordovician Biodiversification Event (GOBE) was one of the most dramatic reorganizations of marine life in Earth history, and it is the necessary ecological backdrop for the argument of this paper. The GOBE is not simply a quantitative increase in the number of species. It is a qualitative reorganization of the structure of marine communities — a shift from the Cambrian-type fauna, dominated by trilobites, inarticulate brachiopods, and archaeocyathids, to a new ecological architecture defined by suspension feeders, complex reef systems, tiered ecological guilds, and increasingly sophisticated predator-prey dynamics.

The numbers are substantial. Marine family diversity roughly tripled between the early Ordovician and the end of the period, with genus-level diversity increasing by a factor of four or more in some analyses. But the number is less important than what it represents. The GOBE produced an ecologically stratified marine community with multiple trophic levels, specialized feeding strategies, elaborate defensive architectures in prey organisms, and — critically — the first major expansion of active, mobile predators capable of sustained pursuit.

The Ordovician reef systems that developed across the tropical and subtropical platforms are particularly relevant here. Tabulate and rugose corals, stromatoporoids (calcareous sponges), bryozoans, and crinoids together constructed carbonate buildups of increasing structural complexity. By the late Ordovician, reef systems on the Laurentian carbonate platform had achieved dimensions and architectural complexity comparable to modern reef ecosystems in terms of their topographic relief, substrate heterogeneity, and ecological stratification. These structures created the physical habitat complexity — the mazes of reef passages, the shadow zones behind buildups, the abrupt bathymetric steps at reef margins — that would have made the spatial structure of predator-prey interaction in these environments qualitatively demanding.

The predators themselves were increasingly capable. Nautiloid cephalopods — relatives of the modern nautilus — were the apex macropredators of the Ordovician ocean. By the Middle and Late Ordovician, orthoconic nautiloids had reached lengths of three to nine meters in some lineages, making them among the largest active predators in the Paleozoic ocean at that time. Their predatory ecology is not perfectly reconstructed, but they were likely sit-and-wait or active pursuit predators targeting soft-bodied and lightly armored prey. They were not the only

predators. Large anomalocaridid relatives persisted into the Ordovician in some lineages. Trilobites, which had diversified extensively in the GOBE, included predatory forms capable of pursuing soft-bodied prey. Sea scorpions (eurypterids) were present and expanding.

For vertebrates, the ecological significance of the GOBE is that it created the predatory environment in which the selective pressure for more capable defense and evasion would have intensified. Jawless vertebrates in the Ordovician occupied a relatively modest ecological position — small filter feeders and perhaps opportunistic predators near the base of the vertebrate size range. But they lived in an ocean that was becoming more dangerous. Prey animals in the Ordovician show clear morphological evidence of escalating defensive investment: thicker shells, more elaborate defensive structures, and greater investment in structures consistent with camouflage and evasion. The arms race between predation pressure and prey defense that defines modern marine ecology was accelerating through this interval.

The ecology of temporally structured environments is also relevant here. Reef systems impose temporal structure on their inhabitants in addition to spatial structure. Tidal cycles, diel light cycles, seasonal productivity cycles, and the irregular dynamics of current flow create a marine environment in which the same location is not the same environment at all times. An organism that could exploit the predictable recurrence of favorable conditions — a refuge opening at low tide, a prey-rich upwelling at a predictable time of day, a thermal refuge at a predictable depth — would have been at an advantage over one that could only respond to present conditions. The GOBE did not invent this problem. It intensified it by making the ecological world more structured, more competitive, and more demanding of precise behavioral timing.

III. OXYGEN, TEMPERATURE, AND OCEAN CHEMISTRY ACROSS THE WINDOW

470–400 Ma

The physical-chemical environment of the ocean across this 70-million-year window was neither static nor uniformly favorable. Understanding its variation is essential for interpreting the selective pressures on vertebrate physiology and behavior, because the metabolic ceiling available to active vertebrates is directly constrained by oxygen availability — and because the disruptions visible in the geochemical record correspond to the extinction and recovery events that are the major evolutionary inflection points of the period.

Atmospheric oxygen levels through the Ordovician are estimated at approximately 13–16% by volume based on carbon and sulfur isotope modeling. This is meaningfully lower than the present 21% and lower than the elevated levels that would characterize the late Devonian and Carboniferous, when atmospheric O₂ may have reached 30–35%. The consequences for vertebrate physiology are significant. Lower atmospheric oxygen implies lower dissolved

oxygen in shallow marine waters, lower metabolic ceilings for aerobic activity, smaller maximum body sizes for active swimmers, and more rapid oxygen depletion in enclosed or poorly-mixed water masses such as lagoons and embayments. Active pursuit predation in the Ordovician was metabolically more expensive per unit of sustained effort than in the modern ocean, which means the evolution of sustained pursuit required not just neural and behavioral innovation but physiological changes that made high-cost activity more viable.

Temperature through the Ordovician was generally warm — most reconstructions place tropical sea surface temperatures at 30–45°C in the early Ordovician, declining somewhat through the Middle and Late Ordovician as the Gondwanan ice sheet developed. This broad warmth supported the carbonate reef ecosystems described in the preceding section, but it also created steep thermal gradients between shallow tropical platforms and deeper or higher-latitude waters. These gradients shaped the distribution of vertebrates and their prey: warm, well-oxygenated tropical platforms were productive and species-rich; deeper or high-latitude environments were colder, less productive, and less favorable for the kind of complex ecological interactions that would generate intense predation pressure.

The ocean's redox structure — the distribution of oxygenated versus anoxic waters across depth and latitude — was substantially different from today. The Ordovician ocean had a larger anoxic interior than the modern ocean. Below the shallow, oxygenated platform seas, anoxic or dysoxic bottom waters occupied much of the deeper ocean basin. This has two implications. First, complex vertebrate life was largely restricted to the oxygenated shallow platforms and shelves — the same environments where reef systems and complex predator-prey ecology were developing. The refuge into deeper water that some modern marine organisms exploit as an escape from surface predation was largely unavailable. The ecological arena was shallower and more spatially constrained than the modern ocean. Second, any perturbation to ocean circulation that expanded anoxic water masses into shelf environments would have been catastrophically destructive to complex marine life — exactly what occurred in the Late Ordovician extinction.

Carbon isotope records from Ordovician carbonates show episodic positive excursions — periods of elevated ^{13}C in seawater carbonate — that correspond to intervals of enhanced organic carbon burial. These excursions, including the major Guttenberg Carbon Isotope Excursion (GICE) at approximately 453 million years ago, reflect disruptions to ocean circulation and productivity that would have caused regional extinctions and restructured communities even before the major end-Ordovician event. The Ordovician ecological world was not a stable background against which evolution proceeded smoothly. It was a periodically disturbed system in which communities were repeatedly restructured, selective pressures shifted with changing oceanographic conditions, and lineages that could persist through disruption and recover ecological function were differentially favored.

Through the Silurian, oxygen levels remained moderate (estimates range from 14–17%) and began a slow rise driven by the diversification of land plants, which began appearing in the

terrestrial record by the mid-Silurian and became increasingly important carbon-buriers through the Devonian. By the Early Devonian, atmospheric oxygen was rising toward 17–19%, and the metabolic ceiling for active vertebrates was correspondingly higher. This timing is not coincidental. The Early Devonian is precisely the interval of the jawed vertebrate radiation — the explosion of placoderm and acanthodian diversity that produced the first ecologically dominant vertebrate predators. The rise in oxygen availability through the Silurian recovery made the evolution of larger, more metabolically active jawed vertebrates more feasible. Physiology and ecology co-evolved.

Ocean temperature through the Silurian recovered from the Hirnantian glaciation and gradually warmed through the period. The Early Devonian was warm — tropical sea surface temperatures may have reached 30–32°C — and the development of the major Devonian reef systems reflects this warmth. The Devonian reef ecosystems, built primarily by stromatoporoids and tabulate corals, represent the most structurally complex marine ecosystems in Paleozoic history before the development of the modern coral reef system. They created the spatial complexity on which the predator-prey ecology of the Devonian fish fauna played out.

IV. THE LATE ORDOVICIAN MASS EXTINCTION (C. 445–443 MA)

Reset, Refugia, and the Restructuring of Selective Pressure

The Late Ordovician Mass Extinction (LOME) is the second-largest mass extinction in Earth history by most metrics, and the first major disruption to the complex marine ecosystem that the GOBE had constructed. It eliminated an estimated 85% of marine species and more than 60% of marine genera over a geologically rapid interval of approximately one to two million years. Understanding it is essential to the argument of this paper, because the LOME was not merely a catastrophe that reset the clock. It was a selective filter whose specific pattern of survival and extinction shaped which lineages would participate in the Silurian recovery — and therefore which lineages would be in a position to exploit the ecological opportunities that recovery created.

The LOME occurred in two pulses, now well-established in the stratigraphic record. The first pulse occurred at the Ordovician-Silurian boundary, approximately 445 million years ago, and was driven by the rapid glaciation of Gondwana. As the Gondwanan ice sheet expanded, global sea level fell by an estimated 50–100 meters, draining the shallow epicontinental seas that had housed the bulk of Ordovician marine diversity. The warm, shallow, highly productive platform environments that had supported reef systems, complex predator-prey ecology, and diverse vertebrate and invertebrate communities were abruptly eliminated across vast areas. What remained was a reduced area of shallower-water habitat, a colder ocean, and a steep drop in global productivity.

The cooling itself was severe. Tropical sea surface temperatures may have declined by 5–10°C across the glaciation interval, with high-latitude cooling even more extreme. Cold water holds more dissolved oxygen than warm water, so the glaciation simultaneously increased dissolved oxygen in surviving shallow-water habitats — a potentially favorable development for the metabolic demands of active vertebrates — while destroying the spatial extent and ecological complexity of those habitats. The net effect was ecological simplification: a smaller, colder, less complex ocean in which the sophisticated predator-prey dynamics of the late Ordovician could no longer be sustained at full scale.

The second extinction pulse occurred approximately one to two million years after the first, as the glaciation ended and sea level rose rapidly. The transgression flooded the platform environments again, but the returning water masses were anoxic — depleted of oxygen by the decomposition of organic material during the lowstand, and poorly mixed as they spread across newly inundated shelves. This anoxic flooding killed off the recovery fauna that had established itself in reduced refugia during the glaciation. The two-pulse nature of the LOME is important: the second pulse selected specifically against the organisms that had survived the first by retreating to refugia, meaning that the survivors of the full event were those with extraordinary ecological resilience, physiological flexibility, or the good fortune to have occupied refugia that persisted through both pulses.

Vertebrates survived the LOME. The jawless fish lineages that had diversified through the Ordovician were reduced but not eliminated. This survival has been variously attributed to the generalist ecology of early agnathans, to their small size (which may have buffered them against food web disruptions), and to their occupation of brackish or nearshore environments that were less affected by the oceanographic changes that devastated open-platform communities. Whatever the mechanism, vertebrate lineages crossed the LOME boundary, and the Silurian world they entered was one in which the ecological guilds formerly occupied by their predators had been selectively devastated. Large nautiloid cephalopods, the dominant apex predators of the Ordovician ocean, were substantially reduced. The complex reef ecosystems that had structured Ordovician predator-prey ecology were largely eliminated. Vertebrates in the early Silurian entered a less dangerous, less complex, and less ecologically structured world — and a world full of ecological opportunity.

The concept of refugia is central to understanding how the LOME shaped subsequent evolution. A refugium, in the paleoecological sense, is a habitat that persists through a major environmental perturbation while surrounding habitats are eliminated. In the context of the LOME, refugia were restricted areas of surviving shallow-water, oxygenated marine habitat — protected lagoons, marginal marine environments, elevated areas of the platform that remained above the falling sea level, and tropical areas where cooling was least severe. These refugia are not ecologically neutral. They are small, isolated, and subject to intense ecological competition among the survivors. Any lineage that persisted in a refugium through the LOME was competing for resources in a compressed ecological space with other survivors, generating intense selection for

ecological efficiency, behavioral flexibility, and — relevantly — the capacity to sustain complex behavioral sequences in a spatially constrained and competitively intense environment.

The LOME therefore did not simply reduce diversity and restart the evolutionary clock. It created a new selective regime for the survivors: small population sizes, restricted geographic ranges, intense intraspecific and interspecific competition in refugia, and the subsequent opportunity to radiate into the ecological vacuum created by the extinction of former community members. This is the specific historical context in which the Silurian recovery — and with it, the jawed vertebrate diversification — occurred.

V. THE SILURIAN RECOVERY (C. 443–419 MA)

Ecological Rebuilding, Vertebrate Opportunity, and the First Jawed Vertebrates

The Silurian period is, from the perspective of the main paper's argument, the most critical interval in the entire chronology. It is during the Silurian that vertebrate diversity begins the expansion that culminates in the Early Devonian radiation of jawed fish. It is during the Silurian that the ecological conditions favorable to the evolution of more capable behavioral control are reconstructed — more slowly, more unevenly, and more complexly than simple recovery narratives suggest. And it is during the Silurian that the first certain jawed vertebrates appear in the fossil record, representing the transition into the substrate class that the main paper proposes as the location of the robust threshold.

The recovery from the LOME was not rapid and it was not uniform. The earliest Silurian (Rhuddanian stage, approximately 443–440 million years ago) shows the lowest diversity values of the post-LOME record — the immediate aftermath of the second extinction pulse. Recovery began unevenly from refugia populations, with different taxonomic groups recovering at different rates and in different geographic regions. Brachiopods, which had survived in higher proportions than many other groups, were among the earliest to diversify. Coral and stromatoporoid reef systems — the critical habitat-structuring elements of the Ordovician — were slow to recover, with fully developed reef ecosystems not reestablishing until the mid-Silurian (Wenlock stage, approximately 433–427 million years ago).

This temporal lag in reef recovery has direct implications for the argument. The mid-Silurian reef resurgence is not merely an ecological curiosity. It represents the reestablishment of the physical habitat architecture — the spatial structure, the substrate heterogeneity, the topographic complexity — that creates the most demanding context for predator-prey ecology. Before the reestablishment of complex reef systems, the marine environment was more ecologically simplified: less structurally complex, lower diversity, lower predation pressure. The recovery of reef systems in the Wenlock marks the point at which the ecological environment began to approach, and eventually exceed, the complexity of the late Ordovician.

The Silurian ocean's oxygen structure gradually improved through the period. Geochemical proxies suggest that anoxic bottom waters retreated somewhat through the Silurian relative to their Ordovician extent, and the reestablishment of platform carbonate ecosystems was associated with improving benthic oxygenation in shallow-water settings. This improving oxygen environment, combined with recovering ecological complexity, set the stage for the metabolically demanding active lifestyles of the Devonian vertebrate fauna.

Vertebrate evolution in the Silurian accelerated markedly relative to the Ordovician. Jawless fish diversified extensively: thelodonts (with their distinctive scale-covered bodies), osteostracans (with distinctive bony headshields), heterostracans, and anaspids all achieve substantial diversity in Silurian deposits. These animals ranged in length from a few centimeters to roughly a meter, and their morphology suggests a range of ecological modes from benthic deposit-feeders to more active swimmers in the water column. The thelodont record in particular shows a diversity of scale morphologies and body forms suggesting ecological partitioning into multiple niches — evidence that vertebrates were beginning to occupy a broader ecological space.

More significant for the paper's argument is the appearance of the first definitive jawed vertebrates. The evidence for jawed vertebrates in the Silurian comes from two main sources: isolated scales and denticles attributable to acanthodians (the earliest jawed fish), and fragmentary remains now recognized as early placoderms. The timing of this appearance — in the Wenlock and Ludlow stages of the Silurian, approximately 430–419 million years ago — places the origin of jawed vertebrates within the context of the recovering, increasingly complex Silurian reef ecosystem. This is not coincidental. The ecological conditions that the recovering Silurian ocean provided — complex spatial structure, recovering predation pressure, expanding ecological opportunity — are precisely the conditions under which the selective advantage of a more capable behavioral control architecture would have been greatest.

The ecology of the Silurian sea scorpions (eurypterids) deserves specific mention. Eurypterids — large arthropod predators related to modern horseshoe crabs — achieved their greatest ecological prominence in the Silurian. Some lineages reached lengths of two meters or more. They inhabited marginal marine and freshwater environments — lagoons, estuaries, nearshore marine settings — that are also among the richest sources of Silurian vertebrate fossils. The co-occurrence of large eurypterid predators with Silurian vertebrates in these marginal marine environments is not accidental. It suggests that the same structurally complex, spatially organized nearshore environments that supported diverse eurypterid predators were also the environments in which vertebrates were evolving under predation pressure. Escape from a large, capable eurypterid predator in a spatially complex lagoonal environment — with substrate features, emergent structures, and geometric constraints on movement — is precisely the kind of behavioral challenge that would reward an organism capable of carrying action context through a multi-step escape sequence.

The vascular plant radiation, beginning in the late Silurian and expanding rapidly through the Devonian, is worth noting in this context. The establishment of terrestrial vegetation created new carbon sinks, began modifying continental weathering regimes, and contributed to the long-term rise in atmospheric oxygen that characterized the later Paleozoic. More immediately, it stabilized river channels and created more complex terrestrial habitats adjacent to the shallow marine environments where vertebrates were diversifying. The terrestrial biosphere was not irrelevant to marine vertebrate evolution — its effects on sediment supply, nutrient runoff, and coastal habitat structure are all documented in the sedimentary record.

VI. THE EARLY DEVONIAN RADIATION (C. 419–393 MA)

Reef Complexity, Jawed Vertebrate Dominance, and the Maturation of Pursuit Ecology

The Early Devonian represents the culmination of the evolutionary processes set in motion by the GOBE, structured by the LOME, and rebuilt through the Silurian recovery. It is the interval in which jawed vertebrates become dominant components of marine ecosystems rather than marginal participants in an invertebrate-dominated world. It is also the interval in which the Devonian reef systems — arguably the most structurally complex marine ecosystems in Paleozoic history — achieve their maximum development. The convergence of these two developments — a diverse and ecologically sophisticated vertebrate fauna operating within the most complex reef ecosystems of the era — creates the environmental and ecological context in which the selective pressure for persistent behavioral control reaches its maximum intensity in the pre-Carboniferous record.

The Devonian reef systems require detailed description because their ecological significance is not captured by the simple observation that reefs existed. The Middle Devonian reef systems of the Canning Basin (Western Australia), the Eifel region (Germany), the Great Lakes region of Laurentia, and the Ural Mountains represent some of the most extensively studied ancient reef ecosystems in the geological record. These reefs were built primarily by stromatoporoids and tabulate corals, with accessory builders including brachiopods, bryozoans, and crinoids. They achieved vertical reliefs of tens of meters in some localities, creating genuine topographic features in the seafloor with all the ecological consequences that topographic structure implies: shadow zones, current-deflected nutrient plumes, shelter from wave action, ambush positions, predator-free refugia in reef interiors, and complex three-dimensional movement corridors.

The ecological structure of these reef systems was layered in space and time. Reef fronts facing open water were exposed to higher current energy and supported fast-growing, wave-resistant corals and stromatoporoids. Reef margins offered vertical faces and overhangs — substrate features that in modern reef systems are used intensively for shelter by prey fish and as hunting corridors by predators. Reef interiors were sheltered, often warmer, and lower in predation

pressure — refugia within the larger reef ecosystem. Reef rubble zones and sand flats associated with reef systems provided further habitat heterogeneity. This is a spatial environment that imposes specific geometry on pursuit-escape dynamics: a predator pursuing a prey animal across a reef environment cannot simply follow the straightest path between two points. Both predator and prey must navigate around structural obstacles, exploit or avoid current acceleration zones, enter and exit cavities, and track a target whose movement is constrained by the same physical features.

The placoderm fauna of the Early and Middle Devonian is the most directly relevant element of the fossil record to the paper's argument. Placoderms were a spectacularly diverse group of jawed fish that dominated Devonian marine and freshwater environments worldwide. By the Early Devonian, placoderm diversity already encompassed multiple ecological modes: small benthic forms feeding on hard-shelled invertebrates, medium-sized active swimmers in the water column, and the first large apex predators in the vertebrate lineage. The arthrodire placoderms — the dominant group — achieved body lengths ranging from under 10 centimeters in small species to over 6 meters in the Late Devonian *Dunkleosteus terrelli*. Even in the Early Devonian, arthrodires of half a meter to a meter in length were present in diversity, occupying predatory niches in the complex reef and nearshore environments.

The jaw mechanism of placoderms is relevant to the behavioral argument. The arthrodire jaw involved a kinetic linkage in which the braincase tilted upward simultaneously with the lower jaw dropping downward, creating an extremely rapid, powerful bite with a very large gape. This mechanism is optimized for capturing fast-moving prey — not for filter-feeding or sediment manipulation. The presence of this predatory anatomy in diverse placoderms by the Early Devonian implies that vertebrate-on-vertebrate predation, and high-speed predatory pursuit, was a significant component of the ecology of Devonian fish communities. Prey fish in these communities were not being approached by passive predators. They were being pursued, intercepted, and captured by fast-moving, capable hunters.

The acanthodians — the other major group of early jawed fish — occupied a different ecological role, generally as small to medium-sized active fish in the water column, feeding on smaller invertebrates and possibly smaller vertebrates. Their diversity in the Early Devonian reef and nearshore environments, combined with their probable schooling behavior (inferred from the frequent occurrence of mass mortality assemblages in the fossil record), suggests that they represented a prey base of considerable abundance. A diverse, abundant prey base of schooling fish operating in complex reef environments — pursued by diverse placoderm predators with sophisticated jaw mechanics — constitutes precisely the kind of ecologically demanding system in which the selective value of persistent behavioral control would have been maximized.

The freshwater record of the Early and Middle Devonian adds a further dimension. Devonian freshwater environments — preserved in the Old Red Sandstone of the British Isles and equivalent formations in eastern North America, Russia, and Australia — contain diverse

vertebrate faunas that include placoderms, acanthodians, and early lobe-finned and ray-finned bony fish. These freshwater environments were not simple streams. They included large river systems, lakes, deltaic marshes, and seasonally variable water bodies. The seasonal variation of freshwater environments imposes temporal structure on their inhabitants in a way that the open marine environment does not: prey availability, water temperature, dissolved oxygen, and habitat geometry change predictably across seasons. An organism that could carry forward organized information about which channel held prey at which season, which refuge was available in low-water conditions, and which competitor used which territory would have been advantaged over one operating only on present conditions. The freshwater Devonian environments, in other words, imposed temporal regularity on their inhabitants in addition to spatial structure — creating exactly the kind of repeated, predictable environmental patterning that rewards persistent internal organization.

VII. REGIONAL VARIATION AND THE UNEVEN GEOGRAPHY OF FAVORABLE HABITATS

A crucial qualification to the narrative developed in the preceding sections is that the ecological environments described were not uniformly distributed across the globe. The selective pressures that the main paper proposes as drivers of controller continuity were concentrated in specific geographic settings — in the structurally complex, well-oxygenated, ecologically rich environments that characterized certain shelf seas and reef zones, not in the entire Ordovician, Silurian, or Devonian ocean. Understanding this spatial unevenness is important because it prevents a misleading reading of the argument: the claim is not that all vertebrates in this interval were under equivalent pressure to evolve durable behavioral control, but that the lineages that eventually gave rise to the first robust continuing controllers were precisely those inhabiting the most demanding ecological environments.

The paleogeographic distribution of Silurian and Early Devonian reef systems is relatively well-constrained. Major reef development was concentrated in the tropical and subtropical belt — within approximately 30 degrees of the paleoequator — where warm, well-oxygenated, carbonate-saturated waters supported the stromatoporoid and coral builders. The Laurentian platform, in tropical latitudes throughout the Silurian and Devonian, was the site of some of the most extensive reef development of either period. The Bohemian terranes (in what is now the Czech Republic) were a second major center of reef development, as were the Carnic Alps region and parts of the North African margin of Gondwana. In contrast, the high-latitude regions of Gondwana — southern Africa, Antarctica, southern South America — were cold, largely glaciated or glacially influenced, and supported different, less structurally complex communities.

The implication is that the most ecologically demanding vertebrate environments — those in which pursuit-escape dynamics in complex reef systems created the strongest selection for

behavioral sophistication — were geographically concentrated. This concentration is reflected in the fossil record: the majority of early jawed vertebrate diversity is found in deposits from the tropical and subtropical platform environments of Laurentia, Gondwana's northern margin, and the Old Red Sandstone continent (the collision zone of Laurentia and Baltica). The vertebrate lineages inhabiting these environments were subject to the full force of the ecological pressures described in the preceding sections. Vertebrate lineages in less complex or less productive environments — the high-latitude margins, the open oceanic settings, the brackish estuaries far from major reef systems — were subject to different, generally less intense, selective pressures.

This geographic concentration also has implications for the timing of the proposed transition. If the evolution of durable controller continuity was driven by the ecological pressures concentrated in these specific environments, then the transition was not a simultaneous global event. It was a localized phenomenon that occurred first in the lineages occupying the most demanding ecological settings, and that subsequently spread through dispersal, radiation, and competitive displacement into less demanding environments. The vertebrate fossil record of the Silurian and Devonian is consistent with this interpretation: early jawed vertebrate diversity is concentrated in the tropical platform environments, and it is from these environments that the later radiation of vertebrates proceeded.

The uneven distribution of habitat quality also structured the distribution of refugia during the environmental perturbations of this interval. The Late Ordovician extinction preferentially eliminated communities in the most exposed, least buffered environments — the open platform settings most vulnerable to sea-level fall, cooling, and anoxic flooding. Refugia persisted in the most sheltered and most ecologically buffered settings: deep marginal basins, protected lagoons, equatorial settings where temperature decline was least severe. These same settings — the most stable, most protected, most spatially complex coastal and lagoonal environments — were also the settings in which the selective pressure for persistent behavioral control would have been most consistent over evolutionary time. Organisms in variable, episodically disrupted environments must evolve behavioral repertoires that can cope with disruption. Organisms in more stable, structurally consistent environments can evolve more specialized, more finely tuned behavioral strategies — strategies that are only advantageous if the environment they are tuned to persists.

The Great Devonian Barrier Reef system of the Canning Basin — now preserved in the Kimberley region of Western Australia and representing one of the best-preserved ancient reef systems in the world — provides a geologically documented example of the kind of specific local environment that would have imposed the most demanding behavioral challenges on its vertebrate inhabitants. This reef system extends for hundreds of kilometers and preserves the full range of reef ecological zones: reef front, reef crest, backreef lagoon, patch reefs, and inter-reef channels. The vertebrate fauna associated with this system includes placoderms and acanthodians characteristic of the ecological context described above. The Canning Basin reef system is not merely an interesting geological feature. It is a preserved record of one of the most ecologically

demanding vertebrate environments of the Early and Middle Devonian, and it is the kind of environment in which the selective arguments of this paper have their most direct application.

VIII. THE CONTROL-ECOLOGICAL ARGUMENT: WHAT THE CHRONOLOGY DEMONSTRATES

The geological and ecological narrative developed in the preceding sections is not a detour from the paper's argument. It is the argument, stated in historical rather than conceptual terms. The main text proposes that persistent selfhood emerged as a control solution to a growing set of ecological and physical demands. This appendix provides the specific historical evidence that those demands were real, that they intensified across the relevant interval, and that they were concentrated in precisely the biological lineages and geographic settings where the main paper locates the proposed transition.

The argument can now be stated in its full temporal form. The world of the Middle Ordovician (470 Ma) was already ecologically complex — the GOBE had produced diverse communities with structured predator-prey dynamics — but the vertebrate fauna was small, jawless, and ecologically marginal. The selective pressure on vertebrates to evolve sophisticated behavioral control existed but was limited by both the vertebrates' ecological role and by the metabolic constraints of lower atmospheric oxygen. The ecological complexity of the Ordovician world was real, but it was an invertebrate-dominated complexity in which vertebrates were not yet the organisms that most urgently needed to remain the same controller through time.

The Late Ordovician extinction (445–443 Ma) disrupted this world but did not simply reset it. It created specific historical conditions — reduced diversity, intense competition in refugia, and the elimination of formerly dominant ecological guilds — that opened the ecological space into which Silurian vertebrates diversified. The extinction was a selective filter, not a random erasure, and the lineages that survived it were those with the ecological flexibility and behavioral resilience to persist through severe environmental instability. These are precisely the characteristics associated with more capable behavioral control: the ability to function across environmental variation, to find resources in reduced or restructured environments, and to maintain effective behavioral sequences despite the disruption of familiar environmental contexts.

The Silurian recovery (443–419 Ma) was the interval in which the ecological world that vertebrates inhabited became, for the first time, genuinely demanding in the specific way that the main paper's argument requires. The reestablishment of reef systems created spatially complex environments. The diversification of eurypterid and early jawed vertebrate predators created intense pursuit-escape ecology in those complex environments. The rise in atmospheric oxygen removed some of the metabolic constraints that had limited the energetic demands of active behavioral sequences. The appearance of the first definitive jawed vertebrates in the late Silurian

occurred in precisely this context — not in an ecologically simple ocean, but in a recovering, increasingly complex, increasingly competitive ecological world in which the value of behavioral sophistication was rising.

The Early Devonian radiation (419–393 Ma) represents the culmination of this trajectory. Diverse placoderms with sophisticated jaw mechanics pursuing diverse acanthodian prey in complex reef environments, with the entire ecology operating in more oxygenated water with more energetic headroom for sustained pursuit — this is the ecological world in which vertebrate behavioral complexity was most strongly selected. It is also the interval in which all vertebrates possess compact myelin, all vertebrates have jaws, and the substrate class proposed in the main paper as the location of the robust threshold is universal in the clade.

The argument from this chronology is therefore not: there was complex ecology, therefore persistent selfhood evolved. The argument is more specific. There was a 70-million-year trajectory of increasing ecological demand on vertebrate behavioral control — interrupted by a major extinction that reset the ecology and created new selective pressures on survivors — and the evolutionary transition proposed in the main paper occurred at the end of this trajectory, in lineages that had been subject to its full force. The timing, the ecology, and the biology align. The emergence of durable controller continuity was not a mysterious event in an empty ocean. It was a plausible, historically grounded response to a world that had spent 70 million years making persistence increasingly worth its material cost.

The chronology presented in this appendix is incomplete in ways that will be resolved by future research. The fossil record of early vertebrates is fragmentary; many intervals and geographic regions are poorly sampled; the detailed ecological relationships among Silurian and Devonian vertebrates can be inferred only indirectly from their morphology and their co-occurrence in sedimentary deposits. The oxygen estimates, temperature reconstructions, and paleogeographic interpretations all carry uncertainty. What is not uncertain is the overall pattern: a trajectory of increasing ecological complexity and predation pressure, structured by specific geographic and bathymetric conditions, punctuated by extinction and recovery events that selectively shaped which lineages survived, and culminating in an ecologically demanding Early Devonian world in which jawed vertebrates with the proposed substrate class were the dominant large-bodied animals in the most complex marine environments on Earth. That pattern is what this appendix is built to document.

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C. Appendix on candidate lineages and comparative substrate classes

PURPOSE: NOT TAXONOMY, BUT SUBSTRATE FUNCTION

This appendix addresses a charge that any substrate-based theory of consciousness evolution must answer directly: that it is engaged in taxonomic chauvinism — singling out a feature of vertebrates and calling it necessary when it is merely the specific solution that one clade happened upon. The charge is not trivial. History is full of theories that mistook the particular for the necessary, and the evolution of complex cognition is an area especially prone to anthropocentric or vertebrate-centric bias.

The argument of this paper is not that compact myelin is the only conceivable solution to the problem of durable controller continuity. It is that compact myelin is the vertebrate way of solving several control problems simultaneously, and that no other biological system that has been examined solves all of them together in a way that produces the substrate class required for the robust continuing controller proposed in Threshold B. The purpose of this appendix is to make that argument explicit, lineage by lineage and criterion by criterion.

The structure of the appendix proceeds from less capable to more capable solutions. Section I establishes the functional criteria — what any substrate must actually achieve to support durable controller continuity. Sections II through V examine the major classes of non-vertebrate and invertebrate solutions, with specific attention to what each buys and what it fails to provide. Section VI examines jawless vertebrates — the lineages that possess vertebrate nervous system organization without compact myelin — as the critical precursor zone. Section VII examines early jawed vertebrates as the proposed location of the first robust Threshold B implementation. Section VIII presents the comparative analysis as a table and as an integrated argument. Section IX addresses what evidence would require revising the proposed threshold location. The analysis throughout uses the neutral, functional language of the main paper: the question is always not which animal is which, but which substrate architecture does what.

I. THE SIX FUNCTIONAL REQUIREMENTS OF A THRESHOLD B SUBSTRATE

Before comparing architectures, it is necessary to be explicit about what the comparison is measuring. The main paper proposes that persistent selfhood requires a continuing controller — one governing organization that can begin an act, revise the act in progress, survive interruption, and return as the same controller rather than a fresh reconstruction. The substrate criterion that follows from this is: an architecture supports durable controller continuity if and only if it can

maintain organized prior state with sufficient fidelity, across sufficient temporal and spatial scale, to govern the behavior of the whole organism through action, revision, and interruption-and-return as one unified process.

This criterion resolves into six specific functional requirements, each corresponding to a real physical constraint on what the substrate must achieve.

The first requirement is speed across distance. As an organism grows in body size and behavioral complexity, signals must travel farther and faster to maintain unified action. If signal propagation speed scales poorly with body size, the organism cannot coordinate actions across its own body within the time windows that complex behavior demands. A predatory strike, an escape maneuver, a mid-course correction: all require that different parts of the body receive and act on coordinated signals within milliseconds. Without sufficient conduction speed, the organism's behavioral control fragments into locally sufficient but temporally uncoordinated responses.

The second requirement is phase preservation. Speed alone is insufficient. For signals to be integrated at downstream neurons, their temporal relationships must be preserved during transmission. A signal that arrives fast but with its timing smeared relative to other signals arriving at the same neuron cannot contribute reliable information to the computation that neuron performs. Phase preservation is what makes it possible for populations of neurons to encode information in the precise timing relationships among their signals — the form of encoding that underlies complex sensory processing, motor planning, and the recursive self-monitoring that the main paper associates with controller continuity.

The third requirement is energy efficiency. Maintaining a continuing controller is a metabolic cost, and that cost is paid continuously — not just during active behavior but during rest, sleep, and inactivity. A substrate that is metabolically unsustainable will be selected against regardless of its functional advantages. The relevant constraint is not merely that the substrate must function during activity but that it must be thermodynamically affordable over the full behavioral cycle of the organism, including extended periods of reduced activity during which the organized prior state must be preserved.

The fourth requirement is metabolic maintenance. This is distinct from energy efficiency. The question is not just how much energy the substrate consumes but whether the substrate includes active mechanisms for maintaining the structural integrity of the conducting elements independent of their firing activity. In the absence of active metabolic maintenance, axons and their associated structures degrade during periods of inactivity. A substrate that cannot maintain its structural organization during sleep or rest cannot guarantee that the controller it instantiates will be preserved through interruption.

The fifth requirement is adaptive plasticity. A continuing controller is not a fixed structure. It must be capable of modification in response to experience, because the organized prior state that governs behavior must be updatable — the controller learns from what it does and what happens

to it. A substrate without plasticity can support only a fixed behavioral repertoire, not the ongoing revision of organized prior state that learning requires.

The sixth requirement is body-wide integration. A continuing controller is an organism-level phenomenon. It must govern the behavior of the whole organism, not just of one region or subsystem. This means the substrate must provide the physical basis for organism-wide coherent signaling — signals from different parts of the body must be integrated within a unified processing architecture, not merely coordinated locally or at the population level.

These six requirements constitute the functional definition of a substrate class that can support Threshold B controller continuity. They are stated in physical terms, not biological terms. Any architecture — biological or synthetic — that meets all six is a candidate substrate class. The following sections examine the major biological architectures against each criterion in turn.

II. NON-VERTEBRATE PARTIAL SOLUTIONS

A. Glial Ensheathment Without Compact Myelin

The most widely distributed form of nervous system support across the animal kingdom is glial ensheathment — the wrapping of axons by glial cells that provide metabolic support, chemical buffering, and some degree of electrical isolation. This form of axon support is ancient and phylogenetically widespread: it is found in annelids (earthworms, leeches), arthropods (insects, crustaceans, arachnids), and other invertebrate phyla. It is also present in the lampreys and hagfish, the jawless vertebrates that predate compact myelin.

Glial ensheathment without compact myelin — where glial cells wrap axons in a loose, non-compacted fashion — achieves several of the functional requirements listed in Section I, but does so incompletely. It provides some metabolic support to axons through glial secretion of nutrients and ions. It provides some electrical isolation that reduces capacitance and improves conduction compared to completely bare axons. It provides physical protection against mechanical disruption. In these respects it is genuinely useful, and its presence across diverse invertebrate lineages reflects genuine adaptive value.

What glial ensheathment without compact myelin does not achieve is saltatory conduction. Without the tightly compacted multilamellar lipid sheath that defines vertebrate compact myelin, and without the clustering of sodium channels at discrete Nodes of Ranvier that saltatory conduction requires, signals propagate continuously along the axon surface rather than jumping between nodes. Continuous conduction is slower (for a given axon diameter), progressively smears phase relationships with distance, and requires ion pumping along the entire axon surface rather than only at nodes. None of these problems is addressed by loose glial ensheathment.

The consequence is that loose glial ensheathment does not achieve criteria one (speed across distance) or two (phase preservation) at the system level required for organism-wide durable controller continuity. It provides a partial solution to criteria three (energy efficiency) and four (metabolic maintenance), but the partial solution does not scale to the demands of a large, behaviorally complex organism. It does not address criterion five (adaptive plasticity) in the specific activity-regulated form that experience-dependent myelination provides. And it does not achieve criterion six (body-wide integration) at the level required for Threshold B.

The clearest examples of the functional limits of ensheathment-only architectures are provided by well-studied invertebrate model organisms. *Caenorhabditis elegans*, with its completely mapped 302-neuron nervous system, achieves impressive behavioral repertoires — chemotaxis, avoidance learning, mate-finding, aggregation — through a fully described neural circuit with no myelin and loose glial support. Its behavioral sophistication is a genuine demonstration of what an ensheathment-only architecture can do. It is not a continuing controller in the sense of the main paper: it cannot revise an ongoing multi-step action in light of mid-sequence information, it has no mechanism for preserving organism-wide organized prior state across interruption, and its behavioral repertoire, while varied, is generated by local circuit dynamics rather than by a unified governing organization that carries prior state through time.

Drosophila melanogaster, with approximately 100,000 neurons and sophisticated glial architecture, represents the upper range of what invertebrate ensheathment architectures achieve. *Drosophila* shows associative learning, circadian rhythms, and complex social behaviors. Its glial cells (in particular, wrapping glia and subperineural glia) provide real metabolic and electrical support to axons. But the *Drosophila* nervous system does not produce saltatory conduction, does not preserve phase relationships across the body scale of even a large insect, and does not provide the kind of organism-wide unified processing that would support Threshold B controller continuity. The fly is behaviorally sophisticated within the limits of its substrate class. Those limits are real.

B. Convergent Multilamellar Insulation in Specific Pathways

A distinct and more directly relevant case is presented by invertebrate lineages that have independently evolved multilamellar (multilayer) insulation around specific axons — typically giant fibers used for escape reflexes. This insulation superficially resembles vertebrate compact myelin in its multilamellar structure, and it has sometimes been cited as evidence that myelin-like insulation is not unique to vertebrates. The comparison requires careful analysis because the functional significance of these convergent structures is substantially different from compact vertebrate myelin.

The best-documented examples are the giant fibers of the penaeid shrimp (*Penaeus*), which show multilamellar wrapping around escape-reflex axons with some resemblance to compact myelin at

the ultrastructural level, and the giant axons of certain annelid lineages (notably the medial giant fibers of earthworms, *Lumbricus terrestris*), which show partial multilamellar ensheathment. In both cases, the insulation is applied to a small number of very large axons dedicated to rapid escape responses — not to the nervous system as a whole.

This localization is the critical limitation. Multilamellar insulation around single giant escape fibers achieves criterion one (speed) for that specific pathway. It does not achieve criterion two (phase preservation) for organism-wide signaling, because it is not present on the vast majority of axons in the animal. It does not achieve criterion four (metabolic maintenance) in the active, activity-independent form that vertebrate oligodendrocytes provide. It does not achieve criterion five (adaptive plasticity) — these structures are not dynamically regulated by neural activity in the way that vertebrate myelin is. And it does not achieve criterion six (body-wide integration), because the vast majority of the nervous system remains uninsulated.

The functional result is an architecture that can produce very fast responses along specific escape-reflex pathways while the rest of the nervous system operates at the slower speed of uninsulated conduction. This is a genuine adaptive solution to the specific problem of rapid escape from predators — it solves a localized, single-pathway control problem with real effectiveness. It does not solve the general problem of organism-wide phase-coherent signaling that durable controller continuity requires. An organism whose escape reflex fires in 5 milliseconds but whose broader behavioral coordination operates at the slower timescales of uninsulated conduction has solved part of the speed problem for one pathway. It has not achieved the substrate class that makes Threshold B possible.

The distinction also matters for evolutionary interpretation. The convergent evolution of multilamellar escape fibers in shrimp and annelids demonstrates that the physics of multilamellar insulation is discoverable by natural selection under the specific pressure of rapid escape-reflex demands. It does not demonstrate that any of these lineages discovered the general solution — the organism-wide, activity-regulated, metabolically maintained, phase-preserving substrate class that vertebrate compact myelin represents. The convergence is real but shallow. It confirms that the underlying physics works; it does not confirm that the full substrate class was achieved.

C. Distributed Local Intelligence: The Octopus as Flagship Stress Test

Distributed local intelligence represents a qualitatively different strategy for solving the control problem — one that does not attempt to achieve organism-wide unified processing at all, but instead distributes processing capacity across semi-autonomous local units that each handle their own region of the body or their own subset of the control problem. The most biologically impressive implementation of this strategy, and the one that poses the most serious challenge to the substrate-class argument, is the cephalopod nervous system — specifically, the octopus.

The octopus nervous system contains approximately 500 million neurons, a figure that places it well above most other invertebrates and in the same order of magnitude as many vertebrates. Of these, roughly 300 million are located in the arms — approximately 35 million per arm, distributed across ganglia running the length of each appendage. The central brain accounts for approximately 170 million neurons, with the remainder distributed between the optic lobes (approximately 40 million per lobe) and other peripheral structures. This distribution is not incidental. It reflects a fundamental architectural decision: the arms are designed to operate with substantial autonomy.

The autonomy is documented directly in the experimental literature. Nesher et al. (2014) showed that octopus arms continue to withdraw from noxious stimuli after surgical disconnection from the central brain — the arm's local neural circuitry is sufficient to generate withdrawal responses without central command. This demonstrates that the arm is not merely a peripheral effector receiving commands from a central controller, but a semi-independent processing unit capable of generating appropriate behavior locally. Hochner and colleagues have shown that the arm's neural circuitry includes local motor programs for complex movements like reaching and fetching that can be initiated and partially controlled by the arm's own ganglia, with the central brain providing initiation signals and high-level direction rather than detailed motor commands.

The behavioral achievements supported by this architecture are genuinely remarkable. Octopuses navigate mazes, open jars, use coconut shells as portable shelters (a form of tool use requiring transport of an object for later deployment), show individual personality differences in boldness and timidity, learn by observing conspecifics performing tasks, and generate extraordinarily sophisticated camouflage patterns involving the simultaneous control of millions of chromatophores — despite being colorblind. The working memory and spatial learning that underlie maze navigation and jar-opening show that the octopus brain maintains organized information about task states and environmental geometry across at least short to medium time windows.

The question this paper must address is specific: does this architecture support Threshold B controller continuity? The answer requires examining not what the octopus can do while active, but what happens to its organized controller state across interruption — and specifically, what the anatomical basis of any such persistence would be.

The case against Threshold B for the octopus rests on four converging lines of evidence. The first is the arm autonomy problem. If the arms are semi-autonomous processing units — if the arm ganglia contain the motor programs for complex arm movements, local sensory processing, and withdrawal responses — then the octopus does not have a single unified continuing controller in the relevant sense. It has a central brain that supervises nine semi-autonomous computational units (one central unit and eight arms). The identity of the organism as a controller is distributed across these units in a way that makes it unclear what would be preserved as a whole across interruption.

The second line of evidence concerns the octopus's sleep behavior. Iglesias et al. (2019) documented striking color pattern changes in sleeping octopuses — rapid, patterned changes of skin coloration appearing during what the authors interpreted as an analog of REM sleep. This finding is often cited as evidence of rich internal processing during sleep, analogous to dreaming. For the purposes of the controller continuity argument, the relevant question is not whether the octopus shows internal processing during sleep, but whether that processing is preserving a unified organism-wide controller state. The skin color changes documented by Iglesias et al. occur in patches and regions that are controlled by local arm and skin ganglia — the same semi-autonomous units that operate during wakefulness. If the color changes reflect local replay in arm and skin ganglia rather than coordinated organism-wide controller state preservation, they are evidence of local maintenance, not of the kind of whole-organism controller preservation that Threshold B requires.

The third line of evidence is the absence of compact myelin. The octopus nervous system, like all cephalopod nervous systems, lacks the compact multilamellar myelin of jawed vertebrates. It has glial cells and some degree of axon ensheathing, but not the organized, activity-regulated, node-structured substrate that achieves all six functional criteria simultaneously. This means that signal propagation in the octopus nervous system, fast as it is in some pathways (aided by very large axon diameters and sophisticated glial support), does not achieve the phase preservation criterion that is required for organism-wide unified processing of the kind that Threshold B requires.

The fourth line of evidence is the octopus's life history. Most octopus species live one to two years. They do not invest in the kind of long-term organizational maintenance that persistent selfhood would require. This is itself a product of selection: if the organism's control architecture does not require durable continuing identity, there is no selective pressure to invest in the metabolic and structural costs of maintaining it. The short lifespan is not a cause but a symptom — it is consistent with an architecture that has not crossed the threshold at which long-term persistence of a unified controller becomes adaptively important.

None of this is a dismissal of octopus cognition. The octopus is among the most remarkable nervous systems in the animal kingdom and challenges many vertebrate-centric assumptions about the relationship between neural architecture and behavioral sophistication. The argument of this paper is not that the octopus is a simple or uninteresting case. It is that the octopus represents the ceiling of what distributed local intelligence can achieve — and that ceiling is below Threshold B. The octopus may reach Threshold A (weak continuing controller): it can maintain action context across brief intervals, it can revise an ongoing manipulation, it shows some evidence of short-term unified behavioral state. But the organism-wide durable persistence of a unified governing organization that Threshold B requires is not supported by the evidence from the octopus nervous system architecture.

The octopus remains, as the main paper describes it, the flagship stress test for the theory. If future research demonstrates that octopus sleep involves coordinated organism-wide state preservation at a level comparable to that of myelinated vertebrates — that what returns after sleep in an octopus is genuinely the same unified controller that entered it — the theory requires major revision. The prediction is that it will not. The test is available. The octopus is the best possible adversarial case, and that is precisely why it must be engaged with the fullest available evidence rather than dismissed.

D. Collective Behavioral Coherence

Collective behavioral coherence — the emergence of apparently unified, intelligent-seeming behavior from the local interactions of many semi-independent individuals or units — requires brief treatment here because it represents the most common false positive in discussions of distributed cognition and the evolution of selfhood. Schools of fish, flocks of birds, insect swarms, and colonial social insects all display behavior that, from a distance, looks like the product of a unified controlling intelligence. None of it is, in the sense relevant to this paper.

The biological mechanisms underlying collective coherence are well-studied. In fish schools and bird flocks, individuals follow simple local rules — maintain a preferred distance from neighbors, align velocity with nearby individuals, avoid collisions — and these rules produce the complex, apparently coordinated group movements that make schools and flocks behaviorally striking. No individual fish in a school knows where the school is going. No individual bird in a murmuration has a plan for the pattern the group is creating. The coherence belongs to the emergent collective dynamic, not to any individual controller.

This point is important for the theory because it addresses a class of argument that might otherwise seem to dissolve the threshold. If sufficiently complex collective behavior is indistinguishable from unified intelligent control when viewed from the outside, why should the presence of a unified internal controller be the relevant criterion? The answer is that the theory is not a theory of behavioral appearances. It is a theory of what must be physically present for the behavior to be generated in the manner relevant to persistent selfhood. A fish school that turns as one does not thereby constitute one continuing controller. The coherence of the group's movement is real. The individual fish's controller continuity — to the extent it exists — belongs to the individual fish, not to the school.

Social insects present a more complex case. An ant colony achieves extraordinary collective intelligence — sophisticated foraging, construction, defense, and fungal farming strategies that are genuinely difficult to explain by simple local rules alone. The colony as a whole shows something like learning, memory, and adaptive decision-making. But the individual ant, not the colony, is the organism with a nervous system. The ant's controller continuity — its capacity to maintain organized prior state across interruption — is what matters for the individual threshold.

The colony's collective intelligence is a remarkable emergent phenomenon that tells us about the possible outputs of distributed coordination; it tells us nothing directly about whether individual ants have crossed the threshold for persistent selfhood as individual controllers.

Collective behavioral coherence therefore functions as a demonstration case for the distinction between coordination and controller continuity. These systems achieve real, sophisticated, sometimes stunning coordination. They do so without any individual component needing to be a robust continuing controller in the Threshold B sense. They are relevant to this paper primarily as a conceptual clarification — the clearest illustration of why "the behavior looks unified" is an insufficient criterion for concluding "there is a unified continuing controller."

III. THE JAWLESS VERTEBRATE PRECURSOR ZONE: LAMPREY AND HAGFISH

Jawless vertebrates — the agnathans, comprising the living lampreys (*Petromyzon marinus* and related species) and hagfish (*Myxini*), as well as the extinct ostracoderms and conodonts that were the dominant early vertebrates of the Ordovician and Silurian — occupy the most critical position in the comparative analysis. They are the organisms that possess vertebrate nervous system organization without vertebrate compact myelin. They are the clearest available biological evidence of what the control architecture of a vertebrate nervous system can achieve before the substrate transition that the main paper proposes as the location of the robust threshold.

The lamprey nervous system has been studied in extraordinary detail, driven partly by the lamprey's accessibility (it is a large, robust organism that survives laboratory conditions) and partly by its position as a phylogenetic outgroup to all jawed vertebrates — any circuit organization found in both lamprey and jawed vertebrates is likely to represent the ancestral vertebrate condition. The resulting literature provides an unusually complete picture of what a pre-myelin vertebrate nervous system can and cannot do.

What the lamprey nervous system achieves is genuinely impressive by invertebrate standards. The lamprey brain contains recognizable homologs of the major brain regions found in all vertebrates. It has a forebrain with olfactory processing and primitive telencephalic regions. It has a midbrain tectum for visual and multisensory integration. It has a hindbrain with the reticular formation, cranial nerve nuclei, and the neural circuitry for respiratory and cardiac control. Most significantly for the paper's argument, it has a basal ganglia circuit whose organization — direct and indirect pathways, dopaminergic modulation, striatopallidal-thalamocortical loops — is essentially homologous to the mammalian basal ganglia at the circuit level (Grillner & Robertson 2016). The basal ganglia are involved in action selection, reward-based learning, and the initiation of voluntary movement — all functions relevant to controller continuity.

The lamprey also has central pattern generators (CPGs) for locomotion that have been studied in detail as models of vertebrate locomotion circuits. These CPGs generate the rhythmic, bilaterally alternating swimming movements of the lamprey through a spinal network of interneurons and motor neurons. Grillner's laboratory has characterized these circuits at the individual neuron level, making the lamprey spinal CPG one of the most completely described motor circuits in vertebrate neuroscience. The CPG can generate swimming movements in an isolated spinal cord, demonstrating that locomotor pattern generation is a local circuit property — but the CPG is normally modulated by descending signals from the brainstem and midbrain that control the speed, direction, and initiation of locomotion.

The lamprey shows conditioning and habituation. It shows social behavior — sea lampreys aggregate in spawning streams, navigate using olfactory cues to return to their natal streams for reproduction (a behavior that requires long-range spatial memory), and show parasitic behavior that involves sustained attachment to host fish over extended periods. The parasitic behavior is particularly relevant: maintaining attachment to a moving, actively resisting host fish over minutes to hours requires the lamprey to sustain a behavioral program — a form of action context — across an extended and variable sequence.

What the lamprey nervous system does not have is confirmed directly by the 2023 lamprey neural cell type atlas published by Lamanna et al. in *Nature Ecology and Evolution*. This study used single-cell transcriptomics to profile the complete repertoire of neural cell types in the lamprey brain, comparing the results to the cell type catalogs of jawed vertebrates. The conclusion relevant to this paper is unambiguous: ancestral vertebrate brains lacked myelinating cells. The lamprey has no oligodendrocytes — the glial cells that produce compact central nervous system myelin in jawed vertebrates — and no cell type corresponding to the Schwann cells that myelinate peripheral axons in jawed vertebrates. The lamprey has glial cells (including radial glia and astrocyte-like cells), but none that produce compact myelin of the vertebrate type.

The functional consequence is that signal propagation in the lamprey nervous system is continuous rather than saltatory. Lamprey axons conduct signals without the speed enhancement, phase preservation, or energy efficiency of saltatory conduction through Nodes of Ranvier. The lamprey does have some large-caliber axons — the Müller cells and Mauthner cell homologs, giant reticulospinal neurons that project from the brainstem to the spinal cord and drive rapid escape responses — whose large diameter provides some conduction speed advantage. But these giant axons are a minority of the total fiber population, and they do not achieve the phase-preservation criterion across the nervous system as a whole.

How does this substrate limitation manifest in the lamprey's behavioral capabilities? The precise answer is difficult to establish from the existing behavioral literature, because the experiments required to test controller continuity directly — the interruption-and-return persistence assays described in the main paper's operationalization section — have not been conducted with the

lamprey as the specific focus. What can be inferred from the available evidence is a pattern consistent with the two-threshold model proposed in the main paper.

The lamprey shows behaviors consistent with Threshold A controller continuity — the weak continuing controller. It maintains parasitic attachment across extended intervals (action context preservation). It navigates back to natal streams using stored olfactory information (organized prior state). Its basal ganglia circuitry provides the neural substrate for action selection based on prior reinforcement history (the functional correlate of organized prior state in the motor-control domain). These behaviors are more than reflexive reactions, and they are more than memory in the minimal sense of retained influence. They suggest an organism that maintains some degree of continuing controller organization across time.

What the lamprey does not show, and what the substrate analysis predicts it cannot show, is the organism-wide phase-coherent signal integration that Threshold B requires. Without myelin, the temporal relationships among signals propagating through the lamprey nervous system degrade with distance in a way that limits the precision and speed of organism-wide coordination. The lamprey nervous system is organized correctly — it has the circuit architecture, the major brain regions, and the neuromodulatory systems of the vertebrate brain — but it is running on substrate that does not achieve the full requirements of the robust continuing controller class.

Hagfish (Myxini) occupy a related but distinct position in the precursor zone. Hagfish are the most phylogenetically basal living vertebrates — or, by some phylogenetic analyses, the sister group to vertebrates rather than vertebrates proper, depending on how the definitional line is drawn. Their nervous systems are substantially simpler than those of lampreys: they lack the well-developed tectum and basal ganglia of the lamprey, and their behavioral repertoire is correspondingly less complex. Hagfish are also confirmed to lack compact myelin. They represent a simpler precursor condition than lamprey — an architecture that may not even reach Threshold A in the same degree that lamprey appears to.

Together, the jawless vertebrates define the precursor zone with precision: nervous system organization that has the basic vertebrate architecture, some degree of continuing controller function consistent with Threshold A, and the absence of the substrate transition that the main paper proposes as necessary for Threshold B. They are the organisms that make the threshold-B argument most testable: if jawless vertebrates can be shown to preserve organism-wide controller state across interruption at a level comparable to myelinated vertebrates, the substrate class argument fails or must be relocated. The prediction is that they cannot and do not. The lamprey neural cell type atlas provides the substrate confirmation. The behavioral literature provides the functional profile. The two together define the precursor zone with a clarity that few other phylogenetic positions in vertebrate evolution achieve.

IV. EARLY JAWED VERTEBRATES: THE FIRST ROBUST SUBSTRATE CLASS

The transition to jawed vertebrates marks the point at which all six functional criteria are met simultaneously for the first time in the vertebrate lineage. This transition is defined, at the substrate level, by the appearance of compact myelin — the tightly compacted, multilamellar lipid sheath produced by oligodendrocytes in the central nervous system and Schwann cells in the peripheral nervous system that enables saltatory conduction through Nodes of Ranvier. All living jawed vertebrates have compact myelin. No living jawless vertebrates have it. The phylogenetic distribution of the substrate class proposed as necessary for Threshold B coincides exactly with the jawed/jawless vertebrate boundary.

The fossil record for the timing of this transition is necessarily indirect — soft tissue does not preserve in most fossil contexts — but genomic evidence provides a complementary line of evidence. Zalc et al. (2008) and subsequent analyses have shown that the key genes for compact myelin formation — including the Myelin Basic Protein (MBP) gene and its regulatory elements, including RetroMyelin sequences (retrotransposon-derived regulatory elements critical for MBP expression) — are present in all jawed vertebrates and absent from jawless vertebrates. The RetroMyelin sequences appear to have been inserted into the vertebrate genome at approximately the time of the jawed vertebrate divergence, suggesting that the genetic machinery for compact myelin production was assembled at or near the jawed/jawless boundary. This is consistent with the paleontological evidence placing the first jawed vertebrates (placoderms and early acanthodians) in the Silurian, approximately 430–420 million years ago, as described in Appendix B.

The earliest jawed vertebrates for which we have substantial evidence are the placoderms — a spectacularly diverse group of armored fish that dominated Devonian aquatic ecosystems worldwide before their complete extinction at the end of the Devonian. The placoderm record does not preserve neural tissue directly, but the inference that they possessed compact myelin is supported by two lines of evidence. First, all other jawed vertebrates possess compact myelin, including the cartilaginous fish (sharks and rays, Chondrichthyes) which are the oldest living myelinated lineage and whose separation from the bony vertebrate lineage predates or is contemporaneous with the placoderms. If the cartilaginous fish have compact myelin, and the lineage diverged from the bony vertebrate lineage before or at the time of the main placoderm radiation, then compact myelin was already present in the common ancestor of cartilaginous and bony fish — which includes the placoderms. Second, the cranial nerve dimensions and neural canal geometries preserved in placoderm skull fossils are consistent with the large myelinated nerve trunks of extant jawed vertebrates, not with the smaller-caliber unmyelinated nerves of jawless vertebrates.

The cartilaginous fish (sharks, rays, and chimaeras) are the oldest living lineage of myelinated vertebrates and provide the most direct evidence for what the substrate class achieves in a

non-mammalian, ecologically representative context. Sharks have been a focus of behavioral research precisely because they challenge assumptions about the relationship between brain complexity and behavioral capability. Several shark species show learning (spatial navigation, associative conditioning), individual recognition, social behavior, and habitat fidelity over multi-year timescales. The lemon shark (*Negaprion brevirostris*) has been shown to return to the same nursery habitat year after year and to recognize individual conspecifics. The Port Jackson shark (*Heterodontus portusjacksoni*) shows spatial learning in maze experiments. These behaviors are consistent with the presence of organized prior state that persists across substantial time intervals — the functional signature of controller continuity.

The ray-finned fish (*Actinopterygii*), which represent the vast majority of living fish diversity, demonstrate the full range of vertebrate behavioral complexity achievable within the myelinated substrate class at non-mammalian brain sizes. Zebrafish (*Danio rerio*), a canonical model organism, show associative learning, social preference, individual recognition, and the classic approach-versus-avoidance decision behaviors noted in the myelin literature as evidence of decision-based rather than purely reflexive behavior (Lopez-Garcia et al. 2024). The cleaner wrasse (*Labroides dimidiatus*) passes the mirror self-recognition test — a behavioral marker that has been used as an indicator of self-awareness in great apes, dolphins, and elephants — in a nervous system an order of magnitude simpler than any mammal's. These findings suggest that the substrate class of compact myelin supports the functional prerequisites for controller continuity across a very wide range of nervous system sizes and complexities.

V. WHY COMPACT MYELIN IS NOT A TAXONOMIC BADGE

The Integrated Solution and What Makes It Distinctive

Having examined both the partial solutions and the proposed substrate class, it is now possible to make the central comparative argument explicit: compact myelin is not a taxonomic badge, and the criterion is not membership in a named clade. The criterion is whether an architecture achieves all six functional requirements simultaneously within a single, unified, organism-wide substrate. The reason compact myelin uniquely meets this criterion — among the biological architectures examined — is that it solves the six problems not as six separate engineering solutions but as one integrated physical system in which the solutions reinforce each other.

The saltatory conduction mechanism achieves criteria one (speed) and two (phase preservation) in a single physical solution. Because action potentials jump between Nodes of Ranvier rather than propagating continuously, signals are regenerated at each node at full amplitude and with precise timing — phase relationships are preserved because each node re-fires at a predictable latency from the preceding node, rather than degrading continuously with distance as continuous conduction does. Speed and phase preservation are not separately engineered solutions; they are two consequences of the same physical mechanism.

The energy efficiency of saltatory conduction (criterion three) is enormous. Ion channels need to be opened only at Nodes of Ranvier — approximately every 1,000 micrometers along a myelinated axon — rather than at every point along the axon surface. The Na⁺/K⁺-ATPase pumps that restore ion gradients after each action potential need to operate only at nodes. Estimates suggest that myelinated conduction requires 100 times less energy per unit distance than continuous conduction in an axon of equivalent caliber — or, equivalently, achieves the same conduction speed in an axon approximately 100 times smaller in diameter, with correspondingly less metabolic investment in axon maintenance. This energy efficiency is not an independent feature; it is the direct metabolic consequence of the same node-restricted action potential mechanism that achieves speed and phase preservation.

The metabolic maintenance function of oligodendrocytes and Schwann cells (criterion four) extends beyond insulation. These cells supply axons with lactate and other metabolic substrates through monocarboxylate transporters (MCTs) expressed at the periaxonal space — the narrow channel between the myelin sheath and the axon membrane. This supply is independent of the axon's firing activity: axons can be metabolically maintained by their myelin-forming cells even during prolonged quiescence. Nave & Werner (2014) showed that oligodendrocyte disruption causes axon degeneration even when conduction is preserved by pharmacological means — demonstrating that the metabolic support function is separable from and in addition to the insulation function. This is the mechanism that makes the controller maintainable through interruption: the organized substrate is actively supported during periods of reduced activity, not merely preserved passively.

Adaptive myelination (criterion five) is the mechanism by which the substrate class incorporates learning and experience. Myelination is regulated by neural activity: axons that fire more are myelinated more thickly, and the internode length and node geometry are adjusted to optimize conduction for the specific firing patterns of each axon. Fields (2015) and subsequent work have documented that this activity-dependent myelination is directly linked to learning and memory formation — blocking myelination impairs learning; enhancing it (in some experimental paradigms) enhances learning. This means the substrate is not static. It is continuously updated by the organism's experience, implementing in the physical structure of the myelin sheath the organized prior state that the controller is carrying forward. The myelin is not just the wire; it is a record of what the wire has been used for.

Body-wide integration (criterion six) follows from the combination of the preceding five. When speed, phase preservation, energy efficiency, metabolic maintenance, and adaptive plasticity all operate together across the organism's full neural architecture — not just along selected escape fibers, not just in one brain region, but throughout the central and peripheral nervous system — the result is a substrate that can support organism-wide coherent processing at the timescales required for unified behavioral control. This is what the non-vertebrate partial solutions cannot achieve: not any one of the six criteria, but all six together, across the whole body.

The argument from integration is the core of why compact myelin is not being treated as a magical taxonomic badge. The relevant question is not "does the organism have a substance called myelin?" It is "does the organism have a substrate that achieves all six functional criteria simultaneously, organism-wide?" The answer happens to correlate perfectly with the presence of compact vertebrate myelin in the biological record, because compact vertebrate myelin is the only biological solution that achieves the integration. Future research might identify other biological or synthetic solutions — but they would have to achieve the same integration, not merely replicate the chemical composition of myelin, to qualify as belonging to the same substrate class.

VI. COMPARATIVE SUMMARY: SUBSTRATE CLASSES AGAINST FUNCTIONAL CRITERIA

Architecture / Lineage	Speed across distance	Phase preservation	Energy efficiency	Metabolic maintenance	Adaptive plasticity	Body-wide integration	Controller verdict
Invertebrate glial ensheathment (worms, <i>Drosophila</i> , cockroach)	Partial	No	Partial	Partial	No	No	Below Threshold A
Convergent multilamellar fibers (shrimp, annelid giant axons)	Yes*	Partial	Partial	No	No	No	Below Threshold A
Distributed local intelligence (octopus, starfish, cnidarians)	Local only	Local only	Local only	No	No	No	Partial Threshold A
Collective behavioral coherence (schools, swarms, slime molds)	N/A	N/A	N/A	N/A	N/A	Group only	False positive (group only)
Jawless vertebrates (lamprey, hagfish)	Partial	No	Partial	Partial	No	Partial	Threshold A (precursor zone)
Jawed vertebrates (placoderms to mammals)	Yes	Yes	Yes	Yes	Yes	Yes	Threshold B (robust continuing controller)

Table C.1. Comparative assessment of major biological substrate architectures against the six functional criteria required for Threshold B controller continuity. Green shading indicates criterion met; yellow indicates partial or context-limited; red indicates criterion not met. The verdict column summarizes each architecture's position relative to the two-threshold model. (*) Fast conduction is achieved only along specific escape-fiber pathways, not organism-wide.

VII. WHAT WOULD MOVE THE THRESHOLD: PREDICTIONS AND FALSIFIABILITY

The comparative analysis developed in the preceding sections generates specific falsifiable predictions about what would need to be observed to move the proposed threshold. Stating these predictions explicitly serves two purposes: it demonstrates that the theory is empirical rather than definitional, and it identifies the specific experimental programs that would most efficiently test it.

The prediction that would most directly falsify the Threshold B location at jawed vertebrates would be a demonstration that lamprey show organism-wide controller state preservation across interruption at a level comparable to myelinated vertebrates. The operationalization of this test is described in the main paper's Section XII: the relevant measurement is persistence of controller-relevant substrate organization across interruption, compared before and after a controlled interruption event (anesthesia, sleep, or equivalent). If lamprey show the same level of persistence as a myelinated vertebrate in this test, then the substrate argument fails, or the threshold must be relocated to a position before the jawed/jawless boundary.

A second prediction concerns the octopus. If the octopus can be shown to preserve an organism-wide unified controller state across interruption — not local arm-level replay during sleep, but coordinated whole-organism state preservation of the kind that would constitute Threshold B — the theory requires major revision. The specific form of this test is whether octopus sleep shows whole-body correlated state preservation or locally distributed, segmental maintenance. The Iglesias et al. (2019) REM-sleep data is suggestive but does not resolve this question at the organism-wide level.

A third prediction concerns the evolutionary timing within jawed vertebrates. If the substrate class is what matters, then the behavioral signatures of controller continuity should appear in the fossil record at the same time as the earliest jawed vertebrates — not in the Carboniferous or Permian, not in the mammalian radiation, but in the Silurian-Devonian boundary where the placoderm record begins. The behavioral evidence for sophisticated action control in Devonian fish is necessarily indirect (morphology and ecological inference rather than direct behavioral observation), but it is consistent with this prediction.

A fourth prediction concerns the comparative neurobiology of the lamprey relative to the simplest myelinated vertebrates. If the substrate class difference is real, there should be measurable differences in the precision and speed of organism-wide neural coordination between lamprey and the simplest jawed vertebrates (such as cartilaginous fish or basal ray-finned fish). These differences should be observable at the physiological level through measures of signal timing precision across the nervous system, and at the behavioral level through the precision and speed of whole-body coordination tasks.

These predictions are not yet tested in the form described here. The experimental program that would test them is technically feasible with current neuroscientific methods — single-unit recording, calcium imaging, and connectomics in lamprey are all established; equivalent methods for octopus are advancing rapidly. The predictions are stated here not as established results but as the specific empirical targets that would confirm or falsify the substrate-class argument. A theory that cannot state such targets is a philosophical position; a theory that states them and exposes them to test is a scientific hypothesis. This theory intends to be the latter.

KEY REFERENCES FOR APPENDIX C

The following selected references underlie the biological claims made in this appendix.

On octopus arm autonomy and distributed processing:

Nesher, N. et al. (2014). Self-learning of grip by the octopus arm. *Current Biology*, 24(12), 1368–1372. Hochner, B. (2012). An embodied view of octopus neurobiology. *Current Biology*, 22(20), R887–R892. Gutfreund, Y. et al. (1996). Organization of the octopus arm. *Journal of Neurophysiology*, 76(3), 1313–1328.

On octopus cognition and learning:

Packard, A. (1972). Cephalopods and fish: the limits of convergence. *Biological Reviews*, 47(2), 241–307. Mäthger, L.M. et al. (2009). Color and pattern analysis in cephalopod camouflage. *Philosophical Transactions of the Royal Society B*, 364, 429–437. Finn, J.K., Tregenza, T. & Norman, M.D. (2009). Defensive tool use in a coconut-carrying octopus. *Current Biology*, 19(23), R1069–R1070.

On octopus sleep and REM-like states:

Iglesias, T.L. et al. (2019). Cyclic sleeping and active skin coloration changes in sleeping octopuses suggest dynamic cognitive processing. *Journal of Experimental Biology*, 222(2), jeb198168. This study documents rapid, patterned chromatophore changes during octopus sleep states and discusses their potential significance.

On lamprey nervous system organization:

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On the lamprey neural cell type atlas:

Lamanna, F. et al. (2023). A lamprey neural cell type atlas illuminates the origins of the vertebrate brain. *Nature Ecology and Evolution*, 7, 1714–1728.
doi:10.1038/s41559-023-02170-1. This paper is the definitive source for the conclusion that ancestral vertebrate brains lacked myelinating cells.

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On the genetics and evolution of compact myelin:

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On convergent multilamellar insulation in invertebrates:

Hartline, D.K. & Colman, D.R. (2007). Rapid conduction and the evolution of giant axons and myelinated fibers. *Current Biology*, 17(1), R29–R35. The definitive review comparing vertebrate myelin with convergent invertebrate solutions and their functional differences.

On collective behavior and local rules:

Couzin, I.D. (2009). Collective cognition in animal groups. *Trends in Cognitive Sciences*, 13(1), 36–43. Sumpter, D.J.T. (2006). The principles of collective animal behaviour. *Philosophical Transactions of the Royal Society B*, 361, 5–22.

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On cartilaginous fish behavior and learning:

Guttridge, T.L. et al. (2009). The role of learning in shark behaviour. *Fish and Fisheries*, 10(4), 450–469. Tricas, T.C. & McCosker, J.E. (1984). Predatory behavior of the white shark *Carcharodon carcharias*. *Proceedings of the California Academy of Sciences*, 43(14), 221–238.

On zebrafish decision-making and behavior:

Lopez-Garcia, C. et al. (2024). Evolution of fish brains and behaviors: how many ways to generate the same outcomes? *Current Opinion in Neurobiology*, 87, 102895.
doi:10.1016/j.conb.2024.102895.

E. Companion paper possibility: a separate methods or theory paper

Finally, the broader project may benefit from a separate companion paper that does what the present paper deliberately avoids doing in its main body. The current manuscript is strongest when it reframes the emergence of persistent selfhood as a control problem and keeps its language neutral, evolutionary, and broadly accessible. A companion paper could then take on the more explicit theoretical and methodological work: naming the full substrate framework, formalizing the proposed thresholds, specifying the persistence measure in greater detail, and laying out the engineering implications for synthetic systems. This separation would be strategically useful. It would allow the present paper to enter the literature as a reframing contribution, while the companion paper would carry the heavier burden of full formalization. In effect, the first paper would change the question, and the second would state, defend, and operationalize the full underlying theory.

Taken together, these appendices and companion sections would allow the project to scale without overloading the main paper. The main manuscript can remain focused on the central question of when organisms first needed to remain the same controller through time, while the appendices preserve the conceptual precision, Earth-system background, comparative biological detail, and experimental vulnerability needed for the larger research program to remain disciplined.